The Gender Gap in Carbon Footprints: Determinants and Implications

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Abstract

Understanding individual carbon footprints is central to designing equitable climate policy, yet little is known about gender differences in emissions. We document that women emit 26 percent less than men from food and transport, using detailed consumption data from France. After accounting for socioeconomic characteristics and differences in the scale of consumption, an 8 percent gap remains. Intra-household dynamics influence the magnitude of the gap, especially for transport. Across household arrangements, residual differences are driven by the disproportionate contribution of red meat and car—but not plane—in men's carbon footprints, two goods that tend to be associated with masculine norms. Our results suggest that climate policies may affect men and women differently.

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1 Introduction

Mitigating climate change requires drastic reductions in carbon emissions and major shifts in consumption behavior (Creutzig et al., 2022), especially in high-emission sectors such as food and transport (IPCC, 2023). A growing literature documents heterogeneity in carbon footprints across income groups (Sager, 2019; Chancel, 2022) and locations (Lyubich, 2025), with important implications for climate policy targeting and equity. Much less is known about systematic gender differences, even though men and women display distinct consumption and mobility patterns. Any gender gap in carbon footprints could reflect differences in socioeconomic factors, climate preferences, as well as social norms and identity-related consumption.[1] Understanding the extent of the gap and its underlying mechanisms is key for climate policy design. If at least part of it arises from differing climate concerns or identity-related consumption choices, climate policies may have gendered implications beyond economic constraints.

This paper quantifies the gender gap in carbon footprints from food and transport, two sectors accounting for about half of individual footprints, and studies the respective contribution of potential drivers.[2] Carbon footprints are defined as the total greenhouse gas emissions (GHG) directly or indirectly caused by individual consumption, measured in tons of CO_2 equivalent (tCO_2e). We link nationally representative individual-level survey data from France to granular emissions intensity measures for over 2,000 food products and car models.[3] Because we observe physical quantities rather than expenditure, we can cleanly separate how much people consume from what they consume.

We begin by documenting the unadjusted gap, defined as the percentage difference in the average carbon footprints of women relative to men. We find that women's carbon footprints from food and transport are, on average, 26 percent lower than men's—a difference comparable to the gap between individuals in households with below-median and above-median income. The gaps are of similar magnitude across both sectors. Our analysis covers 52% of individual carbon footprints (CGDD, 2022).[4] This gap is unlikely to reverse once the remaining sectors—where the literature finds at most a small gender gap—are included (see section 3 for more details).

We then draw on decomposition approaches developed in the gender wage gap literature (Blau and Kahn, 2000) to assess the factors underlying the gap. We organize candidate explanations into those we can speak to directly with our data and those we can only address indirectly. A gender gap in carbon footprints can arise from (i) a composition effect, whereby men and women differ in observable characteristics—income, location, occupation—that are themselves correlated with emissions; (ii) a scale effect, whereby men consume more calories or travel longer distances, partly for biological reasons; (iii) intra-household specialization and bargaining, which allocate consumption and mobility unequally within

[1]Evidence of identity-driven consumption includes Atkin et al. (2021).

[2]In the survey, respondents declare their sex. In the absence of data on gender, we assign to each respondent the gender corresponding to their declared sex.

[3]Our transport data cover any trip, whether the respondent is the driver or a passenger.

[4]Magnitudes are similar for the US: Song et al. (2019).

couples; and (iv) residual differences in product choice at a given scale, whereby men and women select goods of differing carbon intensity. These differences may reflect either greater climate concern among women—as documented in e.g, Bush and Clayton (2023)—or preferences over non-climate attributes that happen to correlate with emissions.

We apply sequential regressions and Gelbach (2016) decompositions to assess the contribution of (i) and (ii) and quantify the unexplained residual. We provide suggestive evidence on (iii) through comparisons across household structures. The residual we associate with (iv) is not separately identified, but the goods in which it concentrates can help distinguish climate concern-driven from other preference-driven explanations.

We find that socioeconomic characteristics explain only part of the unadjusted gap. After adjusting for education, age, household income, type of residence location, and occupation, we find a *characteristics-adjusted gap* of around 20 percent. Scale effects—differences in caloric intake or distance traveled—explain part, but not all, of the gap.[5] For food, the difference is 7 percent once calories are controlled for, leaving 24 percent of the unadjusted gap unexplained. We call this the *scale-adjusted gap*. For transport, the gap narrows to 9 percent after controlling for distance traveled, leaving 35 percent of the original transport gap unexplained.

Comparing singles, couples without children, and couples with children reveals distinct mechanisms behind the gender gap in carbon footprints. For food, women have lower characteristics-adjusted carbon footprints than men across all household structures. For transport, we observe a gap only within couples. It is driven by gender differences in distances traveled that do not exist among singles, suggesting within-household specialization. Yet across all household structures, women's trips are less carbon-intensive per kilometer: conditional on car use, they drive lower-emission vehicles and have higher occupancy rates. The transport gap therefore reflects both intra-household specialization and gender differences in the carbon intensity of travel choices that cannot directly be attributed to intra-household dynamics.

Across the population, the residual gender gap is concentrated in two goods: red meat and cars. Their respective contributions to the residual difference per sector (86 percent for red meat and 100 percent for cars), are far greater than their respective shares in the average individual footprints (14 percent for red meat in food and 82 percent for cars in transport). Both goods are carbon-intensive and closely associated with masculine identity (see e.g., Love and Sulikowski (2018) for meat and Willer et al. (2013) for cars). Men's greater consumption of these goods may reflect lower climate concern or stronger preferences for their non-climate attributes. Strikingly, we find no gender gap in air travel emissions. Like car travel, air travel is a highly emitting mode, but it is less strongly associated with masculine identity. This absence of gap is suggestive of gender differences in preferences that are not only driven by climate concerns.

To assess the broader implications of our findings, we conduct a counterfactual exercise in which

[5] The French public health guidelines (ANSES, 2016) recommend daily caloric intakes of 2,600 kcal for adult men and 2,100 kcal for adult women.

men retain their recommended caloric intake (2,600 kcal per day) and observed travel distances but are assigned women's average carbon intensity (in tCO_2e per calorie or tCO2e per kilometer). Men's carbon footprints would be 3.8 percent lower than they currently are for food, and 16 percent for transport. Such decrease represents, in absolute terms, three times the annual emission reductions expected from the agriculture and transport sectors, respectively, to comply with France's climate targets by 2030 (Haut Conseil pour le Climat, 2024).[6]

To illustrate the policy incidence of the gap we measure, we conduct a back-of-the-envelope calculation. A uniform €100/tCO_2e carbon tax on food and transport would translate into gender-differentiated net burdens once revenues are recycled: about €530 for men and €390 for women annually (2.5% and 1.9% of median disposable income in 2017), assuming that both are tax-inelastic.[7] Assuming equal per-capita recycling, the average net burden would be around +€72 for men and –€68 for women. Although this exercise abstracts from behavioral responses, it highlights that even a uniform carbon tax can generate gender-differentiated outcomes given the large differences in baseline emissions.

Our main contribution is to the literature on heterogeneity in carbon footprints. Beside the well-documented differences across income groups (Chancel, 2022; Sager, 2019) and places (Lyubich, 2025). Some studies have highlighted that men and women differ in the emissions associated with their consumption (Räty and Carlsson-Kanyama, 2010; Toro et al., 2024; Osorio et al., 2024; Arduini and Le Henaff, 2025). These papers usually observe consumption at the household level and infer emissions from expenditure. This makes it difficult to allocate emissions within mixed-gender households and to separate quantities from prices and product quality.[8] These studies leave two questions largely open: whether these gaps persist beyond single-adult households, and whether they reflect differences in quantities consumed, product choices, or household context. We address these questions using representative individual-level data on physical quantities consumed in food and transport.

Second, we connect research on gendered consumption to the literature on gender gaps in economic outcomes (Blau and Kahn, 2017). Studies in economics, psychology, nutrition, and transport document systematic differences between men and women in diet, especially meat consumption (e.g., Rothgerber, 2013; Love and Sulikowski, 2018; Rosenfeld, 2020; Hopwood et al., 2024), and in mobility choices, including commuting, mode choice, and car use (e.g., Wasmer, 2009; Motte-Baumvol et al., 2017; Le Barbanchon et al., 2021). We show that these differences generate sizeable gaps in carbon footprints. Methodologically, we use decomposition tools from the gender wage gap literature to separate the roles of socioeconomic characteristics, consumption scale, and residual differences in the carbon intensity of food and transport choices.

Third, we add to the literature on the distributional and political economy implications of climate

[6] France's reduction targets cover domestic emissions, whereas our consumption-based approach includes imports.

[7] We are not aware of studies estimating differential carbon tax elasticities by gender.

[8] Two products of the same volume can have very different prices yet similar carbon intensity—for example, a 100g premium chocolate and a 100g standard one with similar production processes. Measuring carbon intensity per euro instead of per kilogram, as is common with expenditure data, overestimates the premium chocolate's footprint.

policies such as carbon taxes (Klenert et al., 2018; Douenne and Fabre, 2022; Dechezleprêtre et al., 2025). We highlight gender as a potentially relevant dimension of incidence and support. Even after accounting for socioeconomic characteristics and consumption scale, men consume more high-emission intensity goods than women. The residual gap is concentrated in red meat and cars, but not in air travel. This contrast suggests that the residual is not simply a gender difference in climate concern: it is strongest for goods whose consumption is socially gendered, rather than for carbon-intensive consumption in general. These patterns may translate into gender differences in both the incidence and political support of climate policies.

2 Data and methods

We leverage two surveys, one on food and the other on transport, that we analyze separately. They each record consumption quantities for a representative sample of individuals in the French population. The method for estimating carbon footprints is similar across surveys and consists in multiplying quantities of consumed foods in kilograms (respectively traveled distances in kilometers) by emission intensity at the food product (respectively transport mode or car model) level. Our emission intensity measures reflect the life-cycle greenhouse gas emissions embedded in each final product consumed.

2.1 Building food and transport carbon footprints

2.1.1 Food

We use the INCA3 survey, conducted in 2014–2015 by the French National Health Safety Agency (ANSES). It records food consumption for 2,121 adults. The survey reports daily quantities consumed for around 2,800 food products over three representative days, including food consumed at home and away from home, as well as associated caloric intake. We match these products to life-cycle emission intensities from the Agribalyse database, which covers 2,480 food products consumed in France and accounts for emissions from production, processing, transport, distribution, and cooking. Because the classifications differ across datasets, we match them by combining string matching, NLP, and manual corrections. The full procedure is described in Appendix A. Appendix Figure D.1 shows substantial variation in emission intensity across individual food products, including within broad categories such as meat. Appendix Figure D.2 then aggregates these estimates by food category. Animal meats have the highest average intensities, consistent with previous research (Poore and Nemecek, 2018; Clark et al., 2022). Among them, red meat, defined as ruminant meat excluding veal, has the largest footprint: about three times that of other meat, the next-highest category, which includes cold cuts and mixed meats.

2.1.2 Transport

We use the 2019 French National Transport Survey, which records travel patterns for 12,510 adults. Respondents report all trips below 80 km made on a representative day and all trips above 80 km made over the previous six weeks, covering both daily and long-distance mobility. Trips in personal vehicles are recorded whether or not the respondent is the driver. Trip-level GHG emissions are constructed by the data provider multiplying distances by mode-specific emission intensities.[9] For personal vehicles owned by the household, these intensities are vehicle-specific and based on registration records. We account for upstream manufacturing and vehicle occupancy, with car and two-wheeler emissions allocated across passengers; Appendix A details these adjustments. Appendix Figure D.11 shows that our main results are robust to assigning all car emissions to the respondent irrespective of occupancy.

Appendix Figure D.4 reports average emission intensity by mode, accounting for occupancy and distinguishing short- from long-distance travel. Cars are the highest-emitting mode for short distances, at 168 gCO_2e per km.passenger, while planes are the highest-emitting mode for long distances, at 174 gCO_2e per km.passenger.[10] Compared to the existing literature (e.g Lyubich (2025)), our trip-level emission intensities are more granular, especially for cars, allowing us to study gender differences in the type of car used. Appendix Figure D.3 illustrates this heterogeneity: car-model emission intensities range from under 100 gCO_2e/km to above 300 gCO_2e/km.

2.1.3 Estimation sample

We harmonize individual-level food and transport data to reflect annual carbon footprints per capita for the same population groups.[11] The carbon footprints that we obtain are consistent with per capita carbon footprints estimated for France with top-down approaches combining data on sectoral GHG emissions, trade and input-output tables. Average annual food and transport footprints are 1.9 and 2.7 tCO_2e per person, close to the 2.1 and 2.8 tCO_2e estimates from aggregate French consumption data in Baude (2022). After harmonization, survey-weighted sociodemographic characteristics are similar across the food and transport samples (Appendix Table D.1).

2.2 Quantifying the gender gap in carbon footprints

We first measure the raw gender gap as the percentage difference between women's and men's average carbon footprints. To understand the role of other characteristics, we then estimate adjusted gaps by

[9]For plane, mode-specific emission intensities additionally vary between short- medium- and long-haul flights.

[10]The low per-passenger emission intensity for long-distance car trips reflects higher occupancy rates relative to short-distance car trips. We neglect boats, which have a very low modal share.

[11]We drop individuals aged 80 and more in the transport data since the food survey only interviews individuals until 79. Daily and six-week carbon footprints are scaled to annual values for food and transport, respectively.

sequentially adding groups of covariates to the following regression:

$$Y_i = \beta_0 + \beta_1 Female_i + \beta_2 X_i + \mu_i \tag{1}$$

Y_i is the outcome of interest for individual i, such as her carbon footprint from transport. $Female_i$ is a dummy variable for individual i reporting a female gender. X_i is a vector of socioeconomic, demographic and location controls observed at the individual or household level. μ_i is the error term. β_1 is our coefficient of interest reflecting the association between being female and the outcome. This coefficient reflects the portion of the gender gap that remains after adjusting for observable factors in X_i such as occupational type or income, even though these variables may also be influenced by gender. All averages, decompositions, and regressions use survey weights, so estimates are representative of the national population. We use robust standard errors.

We define the *unadjusted gap* as the estimate from a specification that controls only for survey wave; it closely matches the raw difference in average carbon footprints. Control variables in X_i are harmonized across dataset. Beyond survey wave, we sequentially add controls for age, education, and household size ("+sociodemographics" regression), for rural vs. urban and urban area size ("+location" regression), for categories of household income ("+household income" regression),[12] and employment status (including part-time vs full-time work) and professional category ("+employment status and professional category" regression).[13] Categories are detailed in Appendix B. The gap from this last specification is the *characteristics-adjusted gap*. To examine the role of scale (Sections 4.2 and 4.3), we add calories in the food regression and distances traveled in the transport regression, yielding the *scale-adjusted gap*. Although our regressions yield estimates of the absolute gap, we report our findings as relative gaps by dividing the point estimate on the dummy "female" by the average carbon footprint of men.[14]

We use a Gelbach decomposition to quantify how much each group of covariates contributes to the difference between the unadjusted and adjusted gender gaps, invariant to the order in which covariates are introduced (Gelbach, 2016).[15] This allows for a consistent interpretation of the difference in the estimated gap with and without a given control, even when controls are correlated (Cook et al., 2021). The results should not be interpreted as identifying causal channels, since covariates such as employment, income, location, and commuting distance may themselves reflect gender differences in labor-market outcomes, household responsibilities, and residential choices.

[12] Individual income is not recorded in our data.

[13] A LASSO selection confirmed all controls as relevant, except survey wave in the food survey.

[14] Our main outcome is absolute carbon footprints, so β_1 captures average differences in levels. We use levels rather than logs because 10% of individuals have zero transport emissions, mostly from positive-distance trips by zero-emission modes, complicating log specifications (Chen and Roth, 2024). A log specification yields similar results for the food gap and a larger transport gap than the relative gaps reported in Figure 2a.

[15] The Gelbach decomposition is increasingly used in labor economics (Cook et al., 2021; Bachan and Bryson, 2022) and nests the Oaxaca–Blinder decomposition as a special case.

3 Raw gender gap in carbon footprints

Figure 1 shows the average carbon footprints by gender and consumption category. The annual carbon footprints associated with women's food and transport consumption is 26% lower than men's on average. The gender gap is driven by differences in both food (28% gap, or -0.61 tCO_2e in absolute terms) and transport (25% gap, -0.77 tCO_2e). Despite the right-skewed distributions of carbon footprints (see Appendix Figures D.5, D.6, D.7), the gender gaps are not driven by outliers. After excluding the top and bottom 5% of the distribution within each gender, the gender gap in carbon footprints decreases from 28% to 25% for food. It is unchanged for transport (Appendix Table D.2).

Appendix Figures D.8 and D.9 decompose carbon footprints by consumption context: at-home versus out-of-home food, and work-related versus non-work-related travel, separately for short- and long-distance trips. Work-related travel includes commuting and other business travel, such as client visits. The gender gap in food carbon footprints is larger for food consumed away from home than for food consumed at home: 50% versus 25%. For transport, the gap is driven mostly by work-related travel. The smaller gaps for food consumed at home and leisure-related travel may reflect the role of joint household decisions, which we examine further in section 4.4.

To put our result on the raw gender gap in perspective, we compare it to the income gap in carbon footprints for the same sectors. Splitting the sample into two equal-sized income groups, we obtain a gap of 27%, similar in magnitude to the gender gap and driven by transport. Keeping only the extremes of the distribution—top and bottom income quintiles—yields a gap of 48%, only 1.8 times larger than the gender difference.

Food and transport account for 52% of individual carbon footprints in France. The remaining categories (housing, tangible goods and services) may include sectors in which women have higher emissions. Our estimates therefore do not measure the total gender gap in consumption-based emissions. However, a bounding exercise suggests that omitted sectors would need to exhibit a large reverse gap to offset the food and transport gap: women's footprints in tangible goods and services would have to exceed men's by at least 71% (Appendix C). Clothing, a likely candidate for an offsetting gap, accounts for only 3.5% of household footprints in Europe (European Environment Agency).[16]

[16] Source: 2020 figures https://www.eea.europa.eu/publications/textiles-and-the-environment-the/textiles-and-the-environment-the.

4 Mechanisms underlying the gap

4.1 The role of socioeconomic characteristics

A vast literature documents gender differences in economic conditions, labor market integration, domestic labor, and leisure activities (see Blau and Kahn, 2017, for a review of the gender wage gap). Figure 2a shows the results of sequential regressions obtained in the specifications described in section 2.2. The average gap across the two categories decreases from 26% to 20% after controlling for the full set of household and individual characteristics. Since the 95% confidence intervals of the unadjusted and characteristics-adjusted estimates overlap, we cannot reject that the two are equal.

For food, the gap decreases only slightly, from 28% (-0.61 tCO_2e) to 24% (-0.53 tCO_2e). A Gelbach decomposition highlights that 87% of the food gender gap remains unexplained after adding controls (top panel of Appendix Table D.3). The largest contribution to the difference between the unadjusted and characteristics-adjusted gap comes from employment status and socio-professional category.

For transport, the gender gap narrows from 25% (-0.78 tCO_2e) to 18% (-0.54 tCO_2e) after adding all the controls.[17] The Gelbach decomposition in Appendix Table D.4 shows that location, income, and especially employment status play a role in the gender gap in transport carbon footprints. Thus, the gap comes partly from a composition effect: women are more likely to live in large cities and be part of poorer households, and they are more often unemployed or outside the labor force, all characteristics associated with lower carbon footprints. Still, 70% of the unadjusted gender gap remains unexplained.

Because women's employment rates are lower than men's in France, as in many countries,[18] we re-estimate the unadjusted gender gap among employed individuals. It matches the population gap (28% food, 25% transport) and narrows only slightly—to 26% and 23%—once we add a night-shift control (Appendix Figure D.10).

4.2 Gender differences in caloric intake

The French Health Agency (ANSES, 2016) recommends that women consume on average about 20% fewer calories than men. In our data, however, they consume 27% fewer calories. This excess gap is primarily due to men's higher alcohol consumption, which is not included in dietary guidelines. Thus, part of the food gender gap reflects differences in total consumption beyond biological needs.

[17]The transport survey provides additional controls not available in the food survey, such as detailed occupation, driving license, and the number of cars per household. Appendix Figure D.11 shows that including these factors reduces the gap to 14% but does not eliminate it. The figure also shows that the gap persists when we do not account for car occupancy rate and assign all emissions from car trips to the respondent: the unadjusted gap for transport decreases from 25% to 18% and the characteristics-adjusted gap decreases from 18% to 12%.

[18]In 2019, the gender difference in employment rates was 6 pp in France and 11 pp in the EU on average (Source: Eurostat).[19]

While controlling for recommended intake would better capture biological needs, it is determined entirely by gender and age and is collinear with our sociodemographic controls. We therefore adjust for scale for controlling for observed caloric intake in our regressions. We therefore adjust for scale for controlling for observed caloric intake in our regressions.[20]

Figure 2b shows how the characteristics-adjusted gap decreases when we additionally control for individual caloric intake. The corresponding scale-adjusted gender gap in carbon footprints is 6.7% (-0.15 tCO_2e).[21] This residual represents 24% of the unadjusted gap (see Appendix Table D.3) and reflects systematic differences between men and women in the carbon intensity of the goods consumed.[22]

4.3 Gender differences in distances traveled

The labor and transport literature highlight that men are more willing than women to accept jobs with long commutes (Frändberg and Vilhelmson, 2011; Le Barbanchon et al., 2021). This suggests that, as with food, women's scale of consumption in transport—measured in distance traveled—may be lower than men's. A key difference is that while food intake has a biologically determined upper limit, travel demand is constrained only by time, with faster—and often more carbon-intensive—modes enabling longer distances.[23] This implies a positive correlation between distances traveled and the emission intensity of the transport modes used. Therefore, controlling for the scale of consumption using distances traveled will partial out both the mechanical effect of distances on carbon emissions, and the effect of distance on the transport mode chosen and its intensity.

Given the literature emphasis on gender differences in commuting, we first control for distances traveled coming from commuting exclusively, among the employed. The characteristics-adjusted gap decreases only from 23% to 19% (left panel of Appendix Figure D.12). Part of the gap must, therefore, come from gender differences in the carbon intensity of commuting, or in the scale and carbon intensity of non-commuting trips.[24]

To disentangle the role of scale from emission intensity, we control for total distance traveled (both commuting and non-commuting trips) in the main regression on the full sample. Compared to the characteristics-adjusted gap of 15%, the scale-adjusted gap is of 8.8% (-0.27 tCO_2e, right panel of Figure 2b). While distance appears to be the most important factor explaining the difference between the

[20] Caloric content is not directly linked with carbon footprint; for example, raw sugar is high in calories but low in carbon intensity.

[21] 9.7% if we exclude the calories from alcohol.

[22] The residual share is the scale-adjusted gap divided by the unadjusted gap: $0.147/0.615 \approx 24\%$, equivalently 6.7%/27.7%.

[23] This holds given the low penetration of electric vehicles in France (2.2% of registered cars in 2024 (Ministry for the Environment, 2024)), and the lack of low-carbon alternatives to long-distance flights.

[24] It is unclear whether work-related trips beyond commuting should be included in consumption-based carbon footprints, given their intrinsic link to production activities. The right panel of Appendix Figure D.12 shows that the gender gap in transport carbon footprints persists among the employed when these trips are excluded.

raw and scale-adjusted gaps, accounting for almost half of the overall reduction (bottom panel of Appendix Table D.4), the residual component of 8.8% still represents 35% of the unadjusted gap. This means that conditional on total distances traveled, women's trips are less carbon-intensive than men's, likely due to differences in mode choice or vehicle characteristics. We return to the role of product choice in section 4.5.

4.4 Household arrangements and intra-household bargaining

A large literature documents how intra-household bargaining shapes the distribution of resources and roles within the household (e.g., Almås et al. (2023); Browning et al. (2013)). While these mechanisms are muted in studies of singles only, in our setting we are able to compare the gender gap in carbon footprints across household structures and assess their importance.

Appendix Figure D.13 shows that the unadjusted gender gap exists across all household arrangements: singles, couples without children, and couples with children.[25] Adjusting for controls reveals different patterns for food and transport. Figure 3 shows that the characteristics-adjusted food gap varies little between singles and couples, while the transport gap differs more clearly.

For food, the gender gap is stable across household structures—but this masks a shift in levels: women's predicted food carbon footprint is significantly higher among partnered women without children than among single women, by 9.4% with seasonal and weekday controls and by 5.8% after adding socioeconomic controls (Appendix Table D.5). This asymmetrical convergence pattern is in line with the food studies literature, which finds that women are more likely to align their eating habits to those of their male partners than the other way around (Brown and Miller, 2002; Sobal, 2005; Gregson and Piazza, 2023).

The transport gap decomposes into two margins of emission intensity and distance, that behave differently across household structures (Appendix Figure D.14). Emission intensity per kilometer is significantly lower for women than men in all arrangements, while the gap in distances traveled appears only among couples. The distance gap aligns with intra-household specialization: Blundell et al. (2018) show that couples trade off commuting and childcare, and Le Barbanchon et al. (2021) find that the gender gap in willingness to commute is strongest for married women with children and lowest for single women without. Consistent with this, our transport gap is twice as large for couples with children as without, and the contrast across arrangements comes primarily from work-related trips rather than home-production or leisure trips (Appendix Figure D.15).

A distance gap concentrated among couples could reflect differences in bargaining power and associated resource share. We provide an indirect test for this. Following a literature that uses relative spousal

[25] Single parents are excluded due to the small number of single fathers (N = 16).

education as a proxy for intra-household bargaining power,[26] we classify each respondent in a heterosexual couple by relative educational attainment between partners.[27] Appendix D.16 shows no significant difference in the magnitude of the gender gap in carbon footprints across bargaining-power configurations. This null result suggests that bargaining power, as proxied by relative education, is not the main driver of the distance gap. Moreover, the intensity gap—observed even among singles, where bargaining over household resources is not at play—cannot easily be rationalized by bargaining and instead points to systematic gendered preferences in consumption choices, which we study next.

4.5 Product choice: contributions of red meat, car and plane

Given the residual gender gap in carbon footprints after adjusting for characteristics and scale, men and women must choose products with different carbon intensity. We focus on the three most carbon-intensive goods, red meat for food, and car and plane for transport.

We isolate their contribution to the gender gap by re-estimating equation 1, using as outcomes the annual standardized carbon footprints associated with these goods. Figure D.17 shows the key contributions of red meat and car to the gender gap. Carbon footprints from red meat consumption are 0.38 SD lower for women with only simple controls (unadjusted), 0.34 SD lower when adjusting for all socioeconomic characteristics (characteristics-adjusted), and 0.24 SD lower when additionally adjusting for total caloric intake (scale-adjusted). The corresponding gaps for cars are -0.23 SD, -0.17 SD, and -0.10 SD. The persistent scale-adjusted gap shows that the gap is not only driven by higher calorie requirements and the need to travel longer distances. In contrast, we do not find a significant gender gap for plane emissions, whether considering work-related plane trips or (less constrained) non-work plane trips (Figure D.18).

We further illustrate the role of red meat and cars in Figure 4, where we compare their share in the average carbon footprint with their share in the footprint gap. On average, red meat accounts for 14% of food emissions, while cars account for 82% of transport emissions. We measure their contribution to the gender gap by dividing the coefficient on the gender dummy in regressions where the outcome is red meat (or car)-related carbon footprints by the coefficient in regressions of food (or transport) carbon footprints, using the three specifications presented in Figure D.17. Red meat accounts for 35% of the unadjusted gender gap and 86% of the scale-adjusted gap, a disproportionate share compared to its 14% share in average food emissions. While car already causes 82% of the average transport carbon footprints, gender differences in car emissions explain essentially 100% of the scale-adjusted gap.

We next study the drivers of the gap in car emissions. Unlike red meat, which women consume in smaller quantities relative to men as a share of total food volume, we do not find that women use car significantly less as a share of total distances traveled, once socioeconomic factors are controlled for

[26] Educational attainment is less likely than earnings to be jointly determined within the relationship, as it is often completed before couples form.

[27] Homosexual couples represent 1.1% of couples in the transport data and are excluded from this analysis.

(Appendix Figure D.19). Instead, the gap reflects both shorter car distances and lower emission intensity of women's car trips (Figure D.20). The latter has two components: first, women more often travel with additional passengers. While it can be partly explained by their higher propensity to accompany children (Motte-Baumvol et al., 2017), it is also observed among singles. Second, they drive less carbon-intensive vehicles: in couples with two cars, men are more likely to drive the higher-emission vehicle; among singles, women's cars are substantially less carbon-intensive than men's (-0.4 SD), despite the fact that we find no significant gap in distances traveled between the two.

4.6 Interpreting the residual gap in carbon footprints

We next explore why we observe gender differences in carbon footprints for red meat consumption and car use, but not for air travel. If greater climate concern among women was driving the choice of cleaner goods, we would expect to also find a gender gap in emissions for this good. We can also rule out the role of convergence on leisure plane trips within the household. We find the same pattern—a gender gap in car emissions, but not in plane emissions—when restricting the sample to singles, for whom the intra-household specialization channel can be excluded (Appendix Figure D.21).

Another explanation is that red meat and cars carry risk (e.g., cardiovascular diseases and road accidents) that women, who are more risk-averse (Croson and Gneezy, 2009) try to avoid. For cars we can partly address this: among singles there is no gap in distances driven, the gap coming instead from women owning less carbon-intensive vehicles (Figure D.20). A risk motive would imply women drive less, which we do not observe—and driving smaller cars is not an obvious way to reduce accident risk. More broadly, gender differences in risk-taking may themselves reflect masculinity norms rather than being distinct from them (Matavelli et al., 2026), so this is part of an identity account rather than an alternative to it.

More directly, both red meat and cars are coded as masculine. The social science literature indeed emphasizes the association of masculine identity with red meat consumption (Rothgerber, 2013) and with cars (Scheiner and Holz-Rau, 2012).[28] Masculine identity is a mechanism consistent with all the patterns we document: the residual gap concentrates in red meat and cars, both carbon-intensive and masculine-coded; it is absent for air travel, which is carbon-intensive but a priori not masculine-coded; and the gap in car and red meat emissions persists among singles, for whom we can rule out intra-household dynamics. We therefore view it as likely explanation for the residual gap, while acknowledging that our data do not allow us to measure preferences directly.

[28] For instance, Cuevas et al. (2025) identify many car-related items among the 500 Most Masculine Interests based on Facebook data; random feedback suggesting that men are feminine increases their interest in buying an SUV (Willer et al., 2013); and men are more likely than women to believe that eating meat is natural and necessary for humans (Rothgerber, 2013)

5 Conclusion

In this paper we uncover a significant gender gap in carbon footprints from food and transport in France. Even after adjusting for a rich set of covariates and for differences in the scale of consumption, 24%–35% of the gap remains unexplained, depending on sector. The residual differences we document are influenced by household structure and driven by the disproportionate contribution of red meat and car to men's carbon footprints. The prominence of these high-emission intensity goods associated with masculine identity, but not of plane travel, is consistent with gendered preferences predating climate concerns playing an important role.

Although our analysis focuses on France, our findings likely extend to other high-income countries with similar gender norms and labor market institutions. Gender differences in carbon footprints have been documented in several European countries, albeit with varying methodologies and magnitudes.[29] Other work has documented gender specialization in market and domestic work, gender differences in commuting, and the association of car use and red meat consumption with masculine identities across Western contexts. We show that these same domains account for a substantial share of the observed gender gap. While their quantitative importance may vary across countries, the qualitative patterns highlighted in this paper are unlikely to be specific to France.

Our findings have several implications for climate policy. Our back-of-envelope calculation assuming constant elasticities suggests that men may have a larger carbon tax burden than women in food and transport. Since the perceived individual cost of climate policy strongly shapes support (Dechezleprêtre et al., 2025), women's lower carbon footprints may also translate into greater support for mitigation measures.[30] If the gap partly reflects gendered preferences for carbon-intensive goods, climate policy support could become polarized along gender lines.

[29] Carlsson Kanyama et al. (2021) report a 45% gender gap in transport carbon footprints (excluding holiday car trips and plane trips) for 620 Swedish single-person households, Rippin et al. (2021) find a 33% gender gap in food GHG among 212 UK individuals, and Toro et al. (2024) document a 6.2% difference in emission intensity per euro spent between female- and male-breadwinner households.

[30] Without making it their core research question, some recent studies (e.g., Andre et al. (2025); Den Bijgaart et al. (2026)) report a positive association between female gender and climate policy support or climate donations, conditional on socioeconomic and political characteristics.

References

Almås, Ingvild, Orazio Attanasio, and Pedro Carneiro, *Household decisions and intra-household distributions* 2023.

Andre, Peter, Teodora Boneva, Felix Chopra, and Armin Falk, "Misperceived Social Norms and Willingness to Act Against Climate Change," *Review of Economics and Statistics*, 2025, pp. 1–46.

ANSES, "Actualisation des Repères du PNNS: Révision des Repères de Consommations Alimentaires," 2016.

Arduini, Francesca and Florine Le Henaff, "Female Empowerment and Household Emissions," *Institute for Fiscal Studies Working Paper*, 2025.

Atkin, David, Eve Colson-Sihra, and Moses Shayo, "How Do We Choose Our Identity? A Revealed Preference Approach Using Food Consumption," *Journal of Political Economy*, 2021.

Bachan, Ray and Alex Bryson, "The Gender Wage Gap Among University Vice Chancellors in the UK," *Labour Economics*, 2022, *78*, 102230.

Baude, Manuel, "La Décomposition de l'Empreinte Carbone de la Demande Finale de la France par Postes de Consommation: Transport, Alimentation, Habitat, Équipements et Services," *SDES Working Paper*, 2022.

Blau, Francine D and Lawrence M Kahn, "Gender Differences in Pay," *Journal of Economic Perspectives*, 2000, *14* (4), 75–100.

_ and _ , "The Gender Wage Gap: Extent, Trends, and Explanations," *Journal of Economic Literature*, 2017, *55* (3), 789–865.

Blundell, Richard, Luigi Pistaferri, and Itay Saporta-Eksten, "Children, Time Allocation, and Consumption Insurance," *Journal of Political Economy*, 2018, *126* (S1), S73–S115.

Brown, J Lynne and Daisy Miller, "Couples' Gender Role Preferences and Management of Family Food Preferences," *Journal of Nutrition Education and Behavior*, 2002, *34* (4), 215–223.

Browning, Martin, Pierre-André Chiappori, and Arthur Lewbel, "Estimating Consumption Economies of Scale, Adult Equivalence Scales, and Household Bargaining Power," *Review of Economic Studies*, 2013, *80* (4), 1267–1303.

Bush, Sarah Sunn and Amanda Clayton, "Facing Change: Gender and Climate Change Attitudes Worldwide," *American Political Science Review*, 2023, *117* (2), 591–608.

CGDD, "La Décomposition de l'Empreinte Carbone de la Demande Finale de la France par Postes de Consommation: Transport, Alimentation, Habitat, Équipements et Services," 2022. [Online; accessed 1 Sep. 2025].

Chancel, Lucas, "Global Carbon Inequality Over 1990–2019," *Nature Sustainability*, 2022, *5*, 931–938.

Chen, Jiafeng and Jonathan Roth, "Logs With Zeros? Some Problems and Solutions," *The Quarterly Journal of Economics*, 2024, *139* (2), 891–936.

Clark, Michael, Marco Springmann, Mike Rayner, Peter Scarborough, Jason Hill, David Tilman, Jennie I Macdiarmid, Jessica Fanzo, Lauren Bandy, and Richard A Harrington, "Estimating the Environmental Impacts of 57,000 Food Products," *Proceedings of the National Academy of Sciences*, 2022, *119* (33), e2120584119.

Cook, Cody, Rebecca Diamond, Jonathan V Hall, John A List, and Paul Oyer, "The Gender Earnings Gap in the Gig Economy: Evidence From Over a Million Rideshare Drivers," *The Review of Economic Studies*, 2021, *88* (5), 2210–2238.

Creutzig, F., J. Roy, P. Devine-Wright, J. Díaz-José, F. W. Geels, A. Grubler, N. Maïzi, E. Masanet, Y. Mulugetta, C. D. Onyige, P. E. Perkins, A. Sanches-Pereira, and E. U. Weber, "Demand, Services and Social Aspects of Mitigation," in "Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change," Cambridge University Press, 2022.

Croson, Rachel and Uri Gneezy, "Gender Differences in Preferences," *Journal of Economic literature*, 2009, *47* (2), 448–474.

Cuevas, Angel, Ruben Cuevas, Klaus Desmet, and Ignacio Ortuno Ortin, "The Gender Gap in Preferences: Evidence from 45,397 Facebook Interests," *Journal of Economic Behavior & Organization*, 2025, *238*, 107165.

Dechezleprêtre, Antoine, Adrien Fabre, Tobias Kruse, Bluebery Planterose, Ana Sanchez Chico, and Stefanie Stantcheva, "Fighting Climate Change: International Attitudes Toward Climate Policies," *American Economic Review*, 2025.

Douenne, Thomas and Adrien Fabre, "Yellow vests, pessimistic beliefs, and carbon tax aversion," *American Economic Journal: Economic Policy*, 2022, *14* (1), 81–110.

Frändberg, Lotta and Bertil Vilhelmson, "More or Less Travel: Personal Mobility Trends in the Swedish Population Focusing on Gender and Cohort," *Journal of Transport Geography*, 2011, *19* (6), 1235–1244.

Gelbach, Jonah B, "When Do Covariates Matter? And Which Ones, and How Much?," *Journal of Labor Economics*, 2016, *34* (2), 509–543.

Gregson, Rebecca and Jared Piazza, "Relational Climate and Openness to Plant-Forward Diets Among Cohabitating Couples," *Appetite*, 2023, *187*, 106617.

Haut Conseil pour le Climat, "Tenir le Cap de la Décarbonation, Protéger la Population," 2024.

Hopwood, Christopher J., Jahn N. Zizer, Adam T. Nissen, Courtney Dillard, Andie M. Thompkins, João Graça, Daniela Romero Waldhorn, and Wiebke Bleidorn, "Paradoxical Gender Effects in Meat Consumption Across Cultures," *Scientific Reports*, 2024, *14* (13033), 1–8.

IPCC, "Demand, Services, and Social Aspects of Mitigation. In *Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*," 2023.

Kanyama, Annika Carlsson, Jonas Nässén, and René Benders, "Shifting Expenditure on Food, Holidays, and Furnishings Could Lower Greenhouse Gas Emissions by Almost 40%," *Journal of Industrial Ecology*, 2021, *25* (6), 1602–1616.

Klenert, David, Linus Mattauch, Emmanuel Combet, Ottmar Edenhofer, Cameron Hepburn, Ryan Rafaty, and Nicholas Stern, "Making carbon pricing work for citizens," *Nature Climate Change*, 2018, *8* (8), 669–677.

Le Barbanchon, Thomas, Roland Rathelot, and Alexandra Roulet, "Gender Differences in Job

Search: Trading Off Commute Against Wage," *The Quarterly Journal of Economics*, 2021.

Lezec, Florian, Fabien Perez, and Corentin Trevien, "Calcul des Émissions de Gaz À Effet de Serre dans l'Enquête Mobilité des Personnes 2019," 2023.

Love, Hamish J. and Danielle Sulikowski, "Of Meat and Men: Sex Differences in Implicit and Explicit Attitudes Toward Meat," *Frontiers in Psychology*, 2018, *9*, 307966.

Lyubich, Eva, "The Role of People vs. Places in Individual Carbon Emissions," *American Economic Review*, 2025.

Matavelli, Ieda, Pauline Grosjean, Ralph De Haas, and Victoria Baranov, "Masculinity Norms and their Economic Implications," *Annual Review of Economics*, 2026, *18.*

Motte-Baumvol, Benjamin, Olivier Bonin, and Leslie Belton-Chevallier, "Who Escort Children: Mum or Dad? Exploring Gender Differences in Escorting Mobility Among Parisian Dual-Earner Couples," *Transportation*, 2017, *44* (1), 139–157.

Osorio, Pilar, María-Ángeles Tobarra, and Manuel Tomás, "Are There Gender Differences in Household Carbon Footprints? Evidence From Spain," *Ecological Economics*, 2024, *219*, 108130.

Poore, Joseph and Thomas Nemecek, "Reducing Food's Environmental Impacts Through Producers and Consumers," *Science*, 2018, *360* (6392), 987–992.

Räty, Riitta and Annika Carlsson-Kanyama, "Energy Consumption by Gender in Some European Countries," *Energy Policy*, 2010, *38* (1), 646–649.

Rippin, Holly L, Janet E Cade, Lea Berrang-Ford, Tim G Benton, Neil Hancock, and Darren C Greenwood, "Variations in Greenhouse Gas Emissions of Individual Diets: Associations Between the Greenhouse Gas Emissions and Nutrient Intake in the United Kingdom," *PLOS ONE*, 2021, *16* (11).

Rosenfeld, Daniel L., "Gender Differences in Vegetarian Identity: How Men and Women Construe Meatless Dieting," *Food Quality and Preference*, 2020, *81*, 103859.

Rothgerber, Hank, "Real Men Don't Eat (Vegetable) Quiche: Masculinity and the Justification of Meat Consumption," *Psychology of Men & Masculinity*, 2013, *14* (4), 363.

Sager, Lutz, "Income Inequality and Carbon Consumption: Evidence From Environmental Engel Curves," *Energy Economics*, 2019, *84*, 104507.

Scheiner, Joachim and Christian Holz-Rau, "Gendered Travel Mode Choice: A Focus on Car-Deficient Households," *Journal of Transport Geography*, 2012, *24*, 250–261.

Sobal, Jeffery, "Men, Meat, and Marriage: Models of Masculinity," *Food and Foodways*, 2005, *13* (1-2), 135–158.

Song, Kaihui, Shen Qu, Morteza Taiebat, Sai Liang, and Ming Xu, "Scale, Distribution, and Variations of Global Greenhouse Gas Emissions Driven by U.S. Households," *Environment International*, 2019, *133*, 105137.

Toro, Francisca, Esteban Fernández-Vázquez, and Mònica Serrano, "Who brings emissions home? Comparing female and male breadwinner households by matching techniques," *Energy Policy*, 2024, *190*, 114144.

van Den Bijgaart, Inge, Tommaso Felici, and Jacob Jordaan, "Economic Exposure and Climate

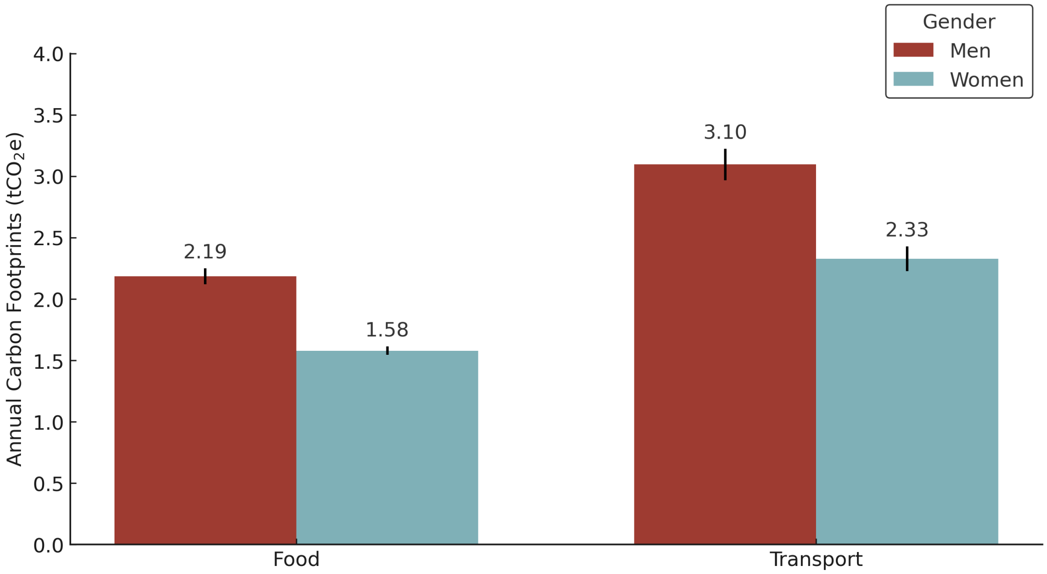
Policy Support," *CESifo working paper*, 2026. [Online; accessed 16. Jun. 2026].

Wasmer, Peter Ruppert & Elena Stancanelli & Etienne, "Commuting, Wages and Bargaining Power," *Annals of Economics and Statistics*, 2009, (95-96), 201–220.

Willer, Robb, Christabel L. Rogalin, Bridget Conlon, and Michael T. Wojnowicz, "Overdoing Gender: A Test of the Masculine Overcompensation Thesis," *American Journal of Sociology*, 2013.

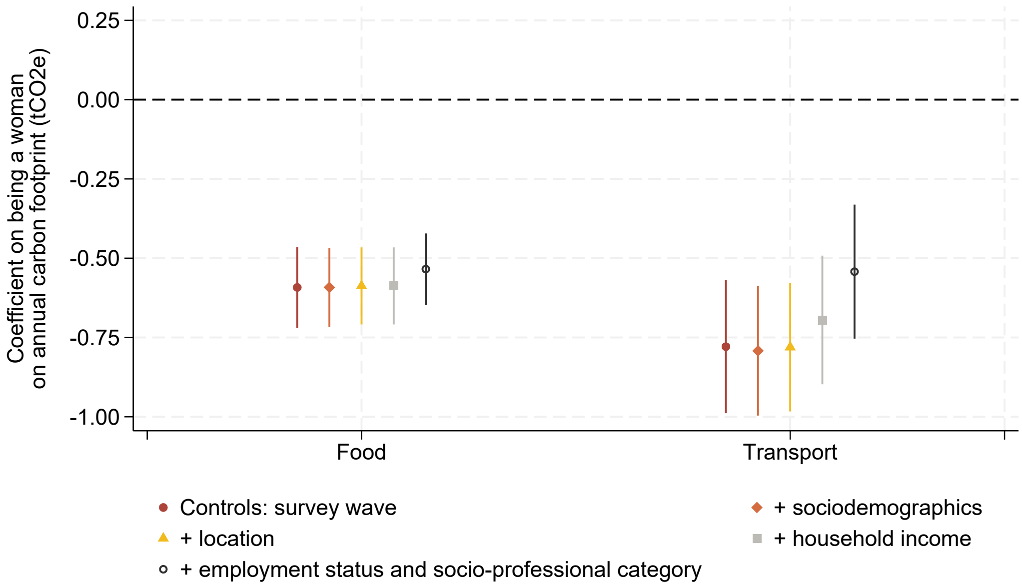
6 Figures

Figure 1: Individual Annual Food and Transport Carbon Footprints by Gender.

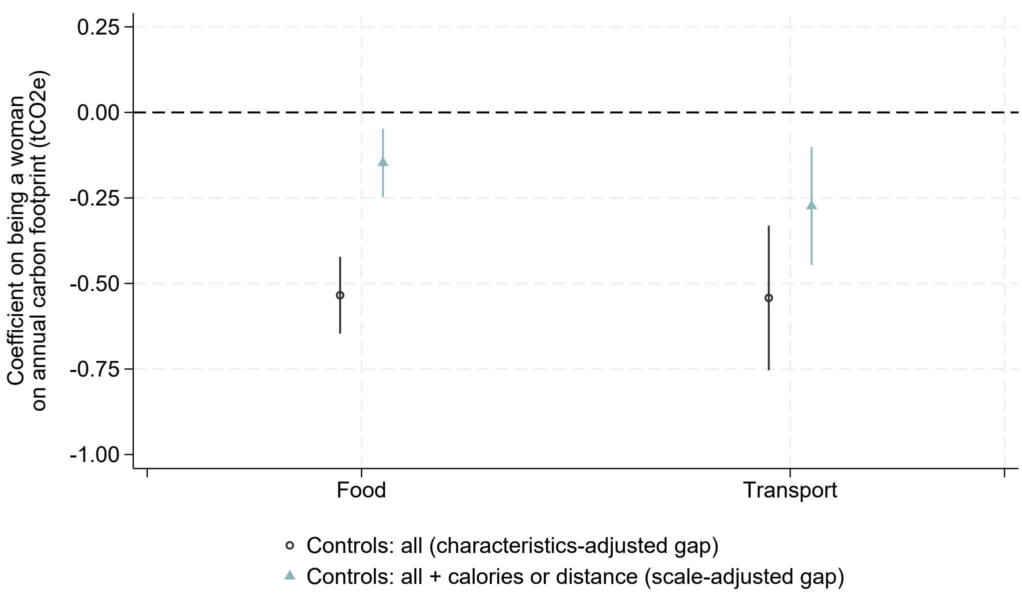


Notes: Sources: Food: INCA3 (N=2,121); transport: EMP (N=11,307). Averages calculated with survey weights. The dark vertical bars indicate 95% confidence intervals.

Figure 2: Gender Gap in Food and Transport Carbon Footprint, the Role of Socio-economic Factors and the Scale of Consumption



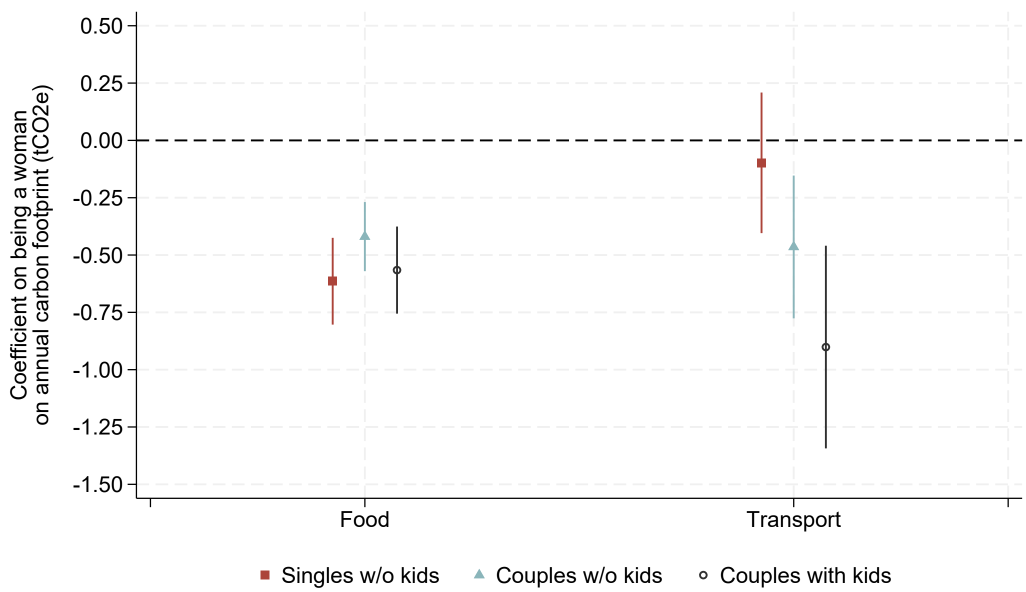
(a) Characteristics-adjusted Gap



(b) Scale-adjusted Gap

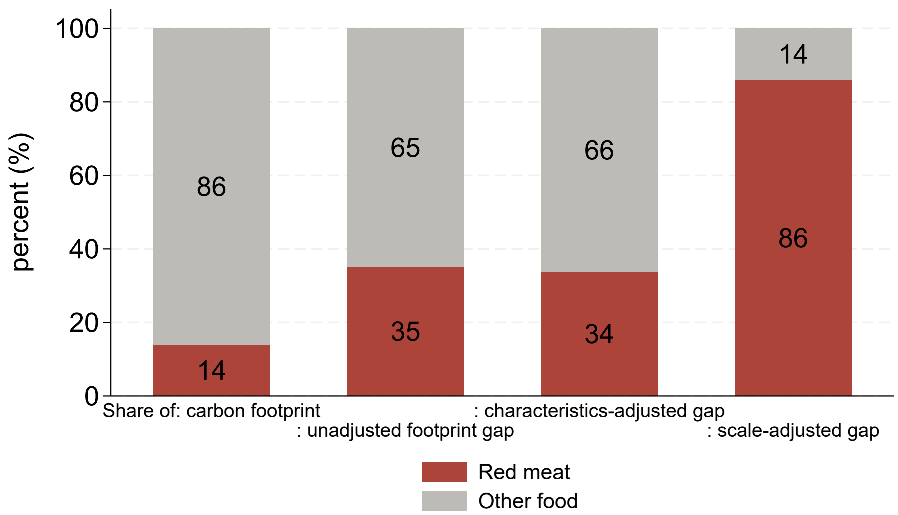
Notes: The point estimates and 95% confidence intervals show the estimated coefficient for the gender dummy "female" from separate OLS regressions, one for each consumption category, with a different set of control variables. In figure (a), "Controls: survey wave" only controls for the time of year when the survey is conducted. "+ sociodemographics" additionally controls for age, education and household size. "+location" additionally controls for rural vs. urban and size of the urban area of residence. "+household income" additionally controls for household income. "+employment status and professional category" additionally controls for employment status and socio-professional category. In figure (b), "Controls: all" controls for survey wave, age, education and household size, location, household income, employment status and socio-professional category and coincides with the last point estimate in figure (a); "Controls: all + calories or distance" additionally controls for caloric intake for food and total distances traveled for transport. Source: Food consumption: INCA3 (N=1,957); transport: EMP (N=11,016)

Figure 3: Characteristics-adjusted Gender Gap in Carbon Footprints by Household Type.

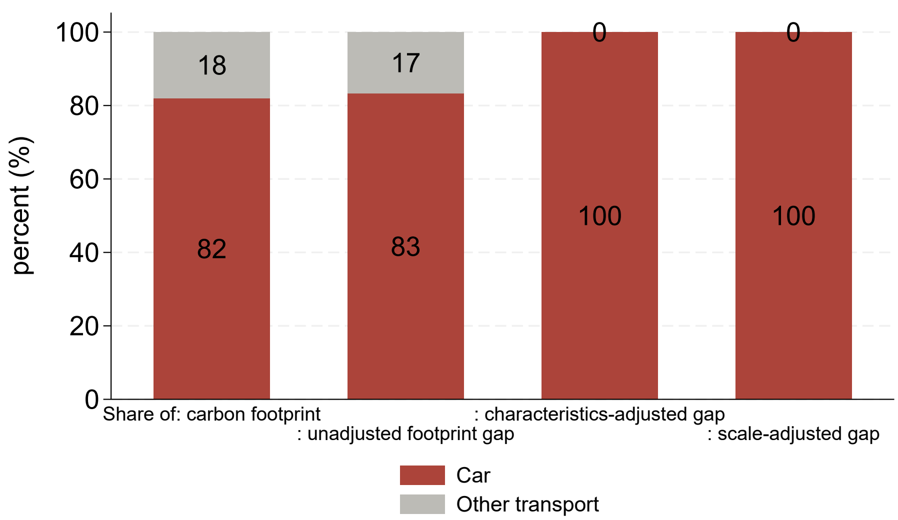


Notes: The point estimates and 95% confidence intervals in red show the estimated coefficient for the gender dummy "female" from separate OLS regressions, one for each consumption category, controlling for the variables listed thereafter, for the subsample of single adults without children. The point estimates, and confidence intervals in blue and black show the corresponding coefficients for the subsample of individuals living in a couple without children and individuals living in a couple with children, respectively. Control variables include the time of year when the survey is conducted, age, education, household size (not for singles), rural vs. urban and size of the urban area of residence, household income, individual employment status and socio-professional category. Source: Food consumption: INCA3 (singles without children: N=545; couples without children: N=721; couples with children: N=610); transport: EMP (singles without children: N=3,490; couples without children: N=3,723; couples with children: N=2,943).

Figure 4: Contribution of Red Meat and Car to the Food and Transport Gaps



(a) Contribution of Red Meat to the Food Gap



(b) Contribution of Car to the Transport Gap

Notes: The first bar shows the relative share of red meat (a) and car (b) compared to respectively other food and other transport in total carbon footprints for the average individual. The second to fourth bars shows the share of the conditional gender gap in food carbon footprints that is explained by the gap in red meat (a) and car (b) emissions. This percentage is obtained by dividing the coefficient on the gender dummy for the regression using red meat (a) and car (b) emissions as outcome by the coefficient on the gender dummy using total food carbon footprints (a) and total transport carbon footprints (b) as outcome. Source: EMP (N=11,016), INCA3 (N=1,957).

Online Appendix

A Methods to Estimate Carbon Footprints

Food Carbon Footprints

The computation of food carbon footprints relies on the matching of food consumption data from the INCA3 dataset and environmental information from the Agribalyse 3.0 dataset, produced by the French Environmental Agency (ADEME). The INCA3 and Agribalyse datasets contain, respectively, 2,886 and 2,481 unique labels that refer to as many standardized products. Given that these datasets have not been matched at the product level before, we rely on a mixed method combining string matching, hand matching and natural language processing (NLP). Doing so, we aim to find for every INCA3 product the closest product in the Agribalyse dataset to associate as precisely as possible food consumption with food environmental impacts. We proceed as follows:

- First, we select perfect string matches defined by a cosine similarity of 1.0. (e.g.,the product label is 'Carot' in both datasets). This is the case for 117 INCA3 products.
- Second, we seek to minimize errors on two types of products: the 263 most consumed products (ie largest volumes per product in the INCA3 dataset), representing together 80% of the purchased volumes; and the top 100 emitting products. For these 363 products, we hand-check the matching and apply hand corrections for one third (118).
- Third, we apply systematic matching based on a mixed method combining NLP and key-term matching.
  - *NLP approach*: we perform NLP matching in two stages. First, we use NLP to identify correspondences between INCA3 and Agribalyse subgroups. Then, within each matched subgroup, we perform a second NLP matching at the product level. The matching is performed using CamemBERT, a deep-learning model trained on French-language text.
  - *Key-term approach*: we define a set of key terms that reflect the most commonly consumed food products in France.[31] This reduces matching error, since the type of ingredient is the key driver of its carbon footprint. In this way, we ensure that the type of ingredient is consistent across datasets.

  For each approach, we compute cosine similarity scores and select the best match. We then compare the matches obtained from the two approaches. In most cases (94% of products), we retain the

[31] We retain a list of 276 key terms which reflect the most common products in the following categories: cheese, dairy, vegetables, fruit, meat, fish, snacks, starches, legumes, drinks, seasonings and culinary aids. Examples of key terms are "broccoli", "parmesan", "bacon"

key-term match rather than the NLP match. The lower performance of the NLP algorithm can be explained by the fact that CamemBERT is not specifically trained on food vocabulary.

- Finally, we perform additional hand-checks for 14% (343) of the products matched in the previous step.

A.1 Transport Carbon Footprints

To reflect real-world lifecycle emissions per person, the following adjustments are made by the data producer to obtain trip-level emissions from transport-specific and car model-specific emission intensities, as explained in Lezec et al. (2023):

- for plane, non-CO_2 warming effects are added
- for cars, emissions associated with cold starts are added. These cold start emissions reflect the fact that the first minutes of a car trip emit more due to the higher fuel consumption of engines until they reach their optimal temperature while driving
- for cars and two-wheelers, the occupancy rate of the trip as declared in the survey is taken into account, and total trip emissions are divided by the number of people in the car/two-wheeler.

We amend these emissions in two ways: first, absent information on distance per mode in multi-modal trips, the calculations assume that the entire trip is done with the main transport mode declared in the survey. We improve the measure by accounting for the distance walked in the trip.[32] Second, we add an emission factor for upstream emissions from vehicle manufacturing using data from the French Agency for the Environment (Base carbone ADEME 2023) so that emissions reflect carbon footprints and are comparable in scope to the food emissions. The only upstream emissions not included are those associated with building the transport infrastructure (roads, rail tracks), due to lack of data.

B Control variables used to estimate the characteristics-adjusted gender gap in carbon footprints

The following control variables are used:

[32] We use trip-level information on the time spent walking and assume a walking speed of 4 kilometers per hour to calculate the distance walked. We proxy the distance traveled with the main transport mode with the difference between total trip distance and walking distance. We re-calculate trip-level emissions by multiplying this distance by the trip-level emission intensity implied by the trip-level emission measure provided in the survey.

- "Controls: survey wave" regression: The variables available differ in the food and transport surveys. For the food survey, we use indicators for the four seasons. For the transport survey, we include the day of the week and two-month sampling periods (e.g., May-June 2018 until March-April 2019). The food survey includes four seasonal indicators: Winter, Spring, Summer, and Fall, while the transport survey uses seven day-of-week and six bi-monthly indicators.

- "+sociodemographics" regression: we control for survey wave and additionally for age, education, and household size. The age categories are 18-44, 45-64, and 65-79 years. Education levels include less than secondary or vocational degree, end of high school diploma, higher education degree ≤ 2 years, and higher education degree > 2 years. Household size is included as a linear control.

- "+location" regression: we control for survey wave, age, education and household size, and additionally for for rural vs. urban and size of the urban area of residence with indicators for the size of the residence's urban area. The categories are: less than 2,000 inhabitants (equivalent to rural), 2,000-19,000 inhabitants, 20,000-99,000 inhabitants, more than 100,000 inhabitants outside the Paris area, and the Paris area.

- "+household income" regression: we control for survey wave, age, education and household size, location, and additionally for household income. We do not have data on individual income or wage. For the food survey, the net income categories range from $<$690€to 4600€per month, with 10 categories. The transport survey uses categories of household income deciles per consumption unit based on the national income distribution. Net income is after transfers and social contributions and before income tax.

- "+employment status and professional category" regression (also referred to as "full set of controls" regression): we control for survey wave, age, education and household size, location, household income, and additionally for socio-professional category and employment status. Socio-professional categories mix activity status and type of occupation for the active individuals (whether employed or unemployed), with the following categories: student, pensioner, other inactive, blue-collar low-skilled, white-collar low-skilled, intermediate occupations, white-collar high-skilled and craftspeople and shopkeepers. For the subsample of individuals in employment, we can further include a dummy variable for whether the individual works night shifts, a characteristic likely to differ between men and women and affect transport options.

- For the subsample of individuals in employment, we further include two variables capturing commuting characteristics in the transport regressions : a continuous measure of commuting distance and a dummy variable for whether the individual works night shifts.

C Estimating a bound on the required gap in footprints in other consumption sectors to cancel out the food and transport gap

Would the gender gap disappear or even reverse if we could observe individual footprints for all consumption categories? Food account for 22% of average footprints in France, and transport for 30% (Baude, 2022). The remaining 48% includes housing (23%), all tangible goods other than food and transport (10%), the reallocation of emissions from final government consumption (8%), and all other services (8%) . Let Gap_{w-m} be the gender gap in carbon footprints across all consumption categories expressed in percent, which takes a negative value if women emit less than men and a positive value otherwise. This gap can be rewritten as the average of category-specific gaps, $Gap_{w-m}^{category}$ weighted by each category's share in total average individual footprint, $S_{CO2}^{category}$. Taking the shares from Baude (2022):

$$\begin{aligned} Gap_{w-m} &= S_{CO2}^{food} Gap_{w-m}^{food} + S_{CO2}^{transport} Gap_{w-m}^{transport} + S_{CO2}^{housing} Gap_{w-m}^{housing} + S_{CO2}^{gov} Gap_{w-m}^{gov} + S_{CO2}^{other} Gap_{w-m}^{other} \\ &= 0.22 Gap_{w-m}^{food} + 0.3 Gap_{w-m}^{transport} + 0.23 Gap_{w-m}^{housing} + 0.08 Gap_{w-m}^{gov} + 0.18 Gap_{w-m}^{other} \end{aligned}$$

Where *gov* stands for government consumption, and *other* includes all goods and services other than food, transport, housing and government consumption.

Based on our estimates, $Gap_{w-m}^{food} = -0.28$ and $Gap_{w-m}^{transport} = -0.25$. Housing emissions are hard to assign within the household, so the only proxy we have on gender gaps is for singles: a study in four European countries reports unadjusted gender gaps in housing energy consumption between -4% (women lower than men) and +4% (women higher than men) depending on the country (Räty and Carlsson-Kanyama, 2010). Assuming that the gap for France lies in this interval, that men and women have a similar carbon intensity for housing energy, and that singles are representative of the overall population, this gives a housing footprint gap of at most +4% with lower carbon footprints for men, or $Gap^{housing} = 0.04$. For government consumption, the dominant approach in the literature is to allocate equal emissions to each individual, so $Gap^{gov} = 0$. We obtain:

$$\begin{aligned} Gap_{w-m} &= 0.22 \times (-0.28) + 0.3 \times (-0.25) + 0.23 \times (0.04) + 0.08 \times 0 + (0.10 + 0.08) \times Gap_{w-m}^{other} \\ &= 0.18 \times Gap_{w-m}^{other} - 0.1274 \end{aligned}$$

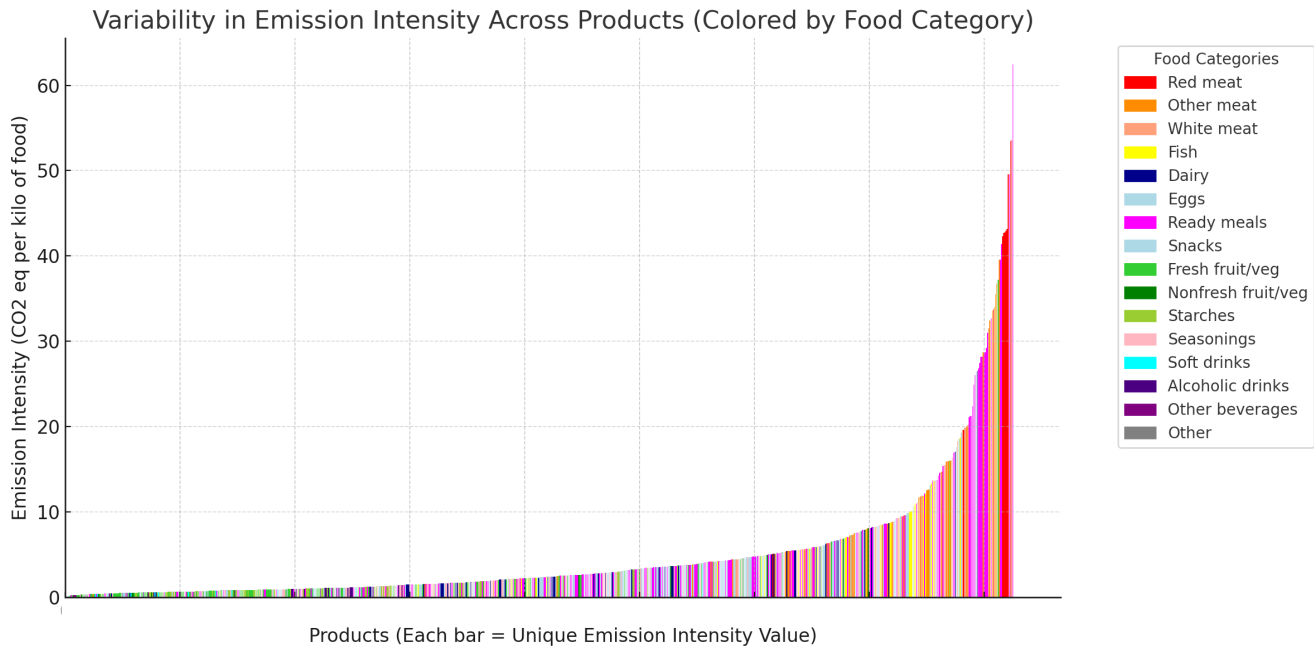
And so we have a positive overall gap, meaning a larger footprint for women than for men, if and only if:

$$Gap_{w-m} > 0 \Leftrightarrow Gap_{w-m}^{other} > \frac{0.1274}{0.18} \Leftrightarrow Gap_{w-m}^{other} > 0.708$$

D Additional Figures and Tables

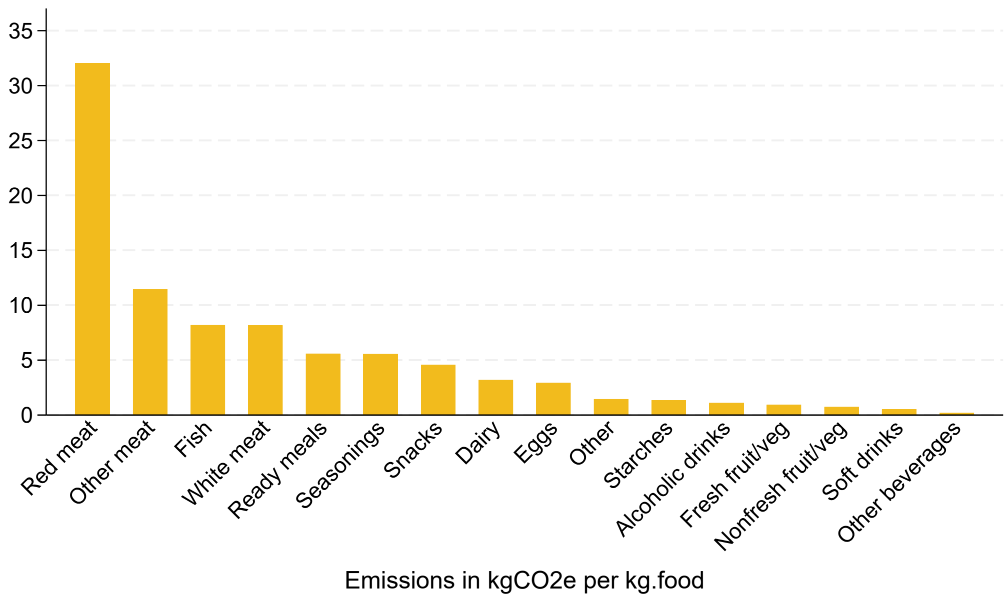
D.1 Figures

Figure D.1: Food Emission Intensity by Product Type.



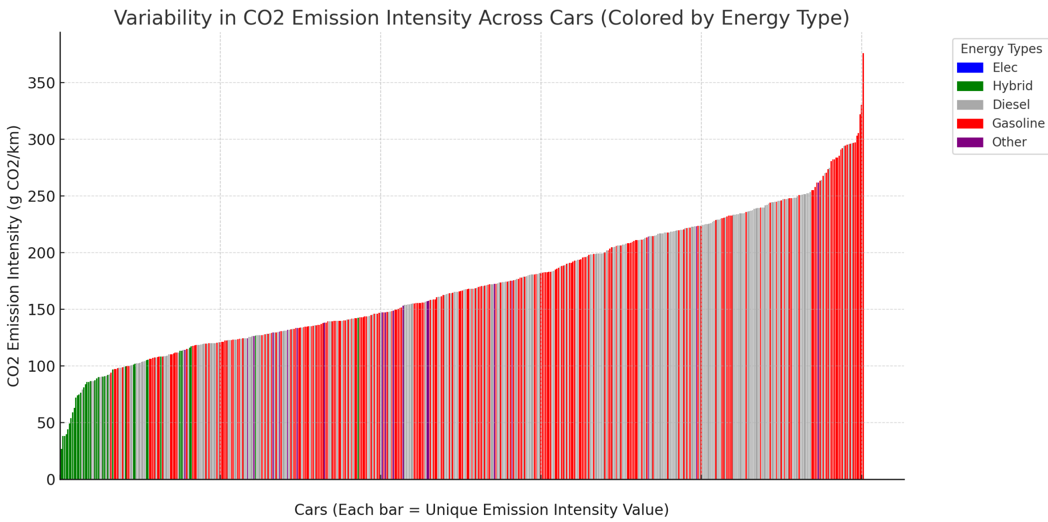
Notes: Distribution of greenhouse gas (GHG) emission intensities (in CO_2e per kilogram of food) across all food products. Each bar represents a *unique value* in the total distribution of emission intensities, sorted from lowest to highest. The visualization highlights the granularity and variability in climate impacts among different food items. Food categories are defined as follows: *Starchy food*: pasta, bread, semolina, cook-type cereals, potatoes; *Fresh fruits and vegetables*: fresh fruits, fresh vegetables; *Red meat*: beef, mutton, lamb; *White meat*: chicken, turkey, veal, rabbit, pork, poultry; *Other meat*: cold cuts, mix of meats, ham, game meat, frogs, kangaroo; *Fish*: fish and seafood; *Eggs*: hen and quail eggs; *Dairy*: cheese, milk, yogurts; *Snacks*: sugary biscuits, jam, honey, spreads, cereal/granola bars, chocolate, pastries, breakfast drink preparation, breakfast cereals, ice cream, desserts, dry fruits and seeds, salty biscuits, olives, crisps; *Ready meals*: prepared dishes, frozen dishes, canned dishes; *Soft drinks*: sodas, syrups, juices; *Alcoholic drinks*: cocktails, liquors, wine; *Other beverages*: coffee, tea, infusions, water, chicory; *Seasonings*: spices, oil, vinegar, butter, croutons, breadcrumbs, raw pastry, confectionery flavors, flour, prepared crust, coconut milk, cream, lemon juice, herbs, dry herbs, garlic, onion; *Non-fresh fruits and vegetables*: canned fruits and vegetables, frozen fruits and vegetables, lyophilized vegetables, beans, dried vegetables, packaged vegetables; *Other*: baby food, chewing gum, food supplements. Source: INCA3 food intake-by-day data (N=256,301). Weighted averages across food intakes of the same food category and day, using food quantities as weights. Source: Agribalyse (2017).

Figure D.2: Average Emission Intensity by Food Category.



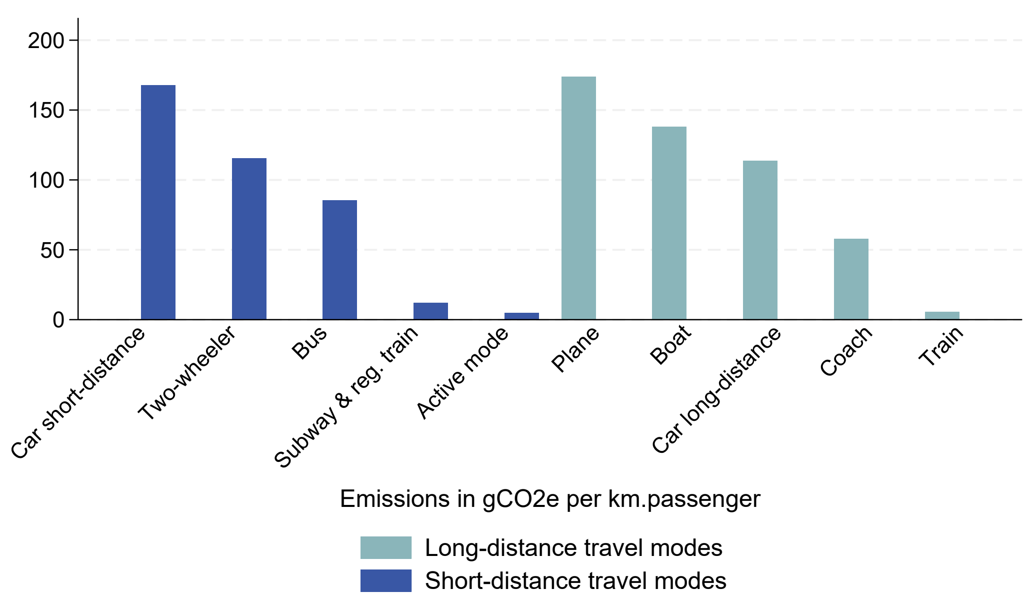
Notes: Volume weighted CO_2e emissions in kilograms expressed in kilogram of the food consumed. Weighted averages across food intakes of the same food category and day, using food quantities as weights. Food categories are defined as follows: *Starchy food*: pasta, bread, semolina, cook-type cereals, potatoes; *Fresh fruits and vegetables*: fresh fruits, fresh vegetables; *Red meat*: beef, mutton, lamb; *White meat*: chicken, turkey, veal, rabbit, pork, poultry; *Other meat*: cold cuts, mix of meats, ham, game meat, frogs, kangaroo; *Fish*: fish and seafood; *Eggs*: hen and quail eggs; *Dairy*: cheese, milk, yogurts; *Snacks*: sugary biscuits, jam, honey, spreads, cereal/granola bars, chocolate, pastries, breakfast drink preparation, breakfast cereals, ice cream, desserts, dry fruits and seeds, salty biscuits, olives, crisps; *Ready meals*: prepared dishes, frozen dishes, canned dishes; *Soft drinks*: sodas, syrups, juices; *Alcoholic drinks*: cocktails, liquors, wine; *Other beverages*: coffee, tea, infusions, water, chicory; *Seasonings*: spices, oil, vinegar, butter, croutons, breadcrumbs, raw pastry, confectionery flavors, flour, prepared crust, coconut milk, cream, lemon juice, herbs, dry herbs, garlic, onion; *Non-fresh fruits and vegetables*: canned fruits and vegetables, frozen fruits and vegetables, lyophilized vegetables, beans, dried vegetables, packaged vegetables; *Other*: baby food, chewing gum, food supplements. Source: INCA3 food intake-by-day data (N=256,301).

Figure D.3: Cars Emission Intensity by Fuel Type.



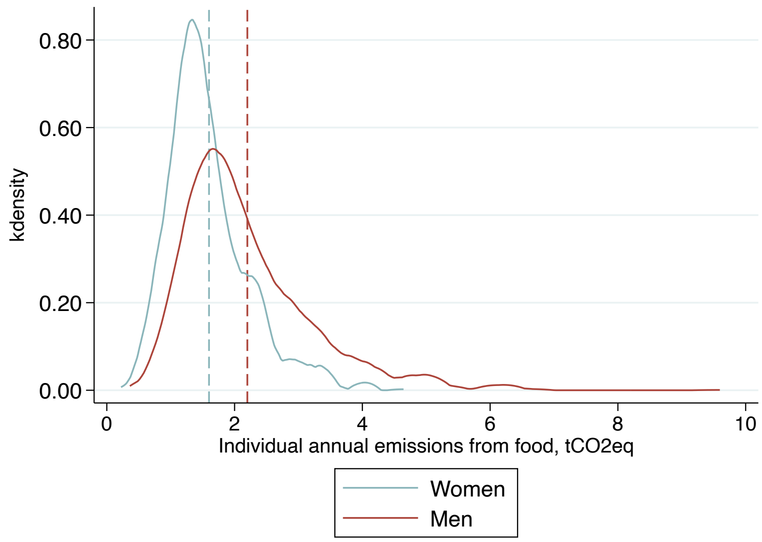
Notes: Distribution of CO_2e emission intensities (in g CO_2e/km) for cars. Each bar corresponds to a *unique value* in the total distribution of cars emissions, sorted from lowest to highest, rather than to an individual vehicle. Bars are colored according to the vehicle's energy type associated with each intensity value. *Other* represents liquefied petroleum gas cars. Source: EMP data.

Figure D.4: Average Emission Intensity by Transport Mode Category.



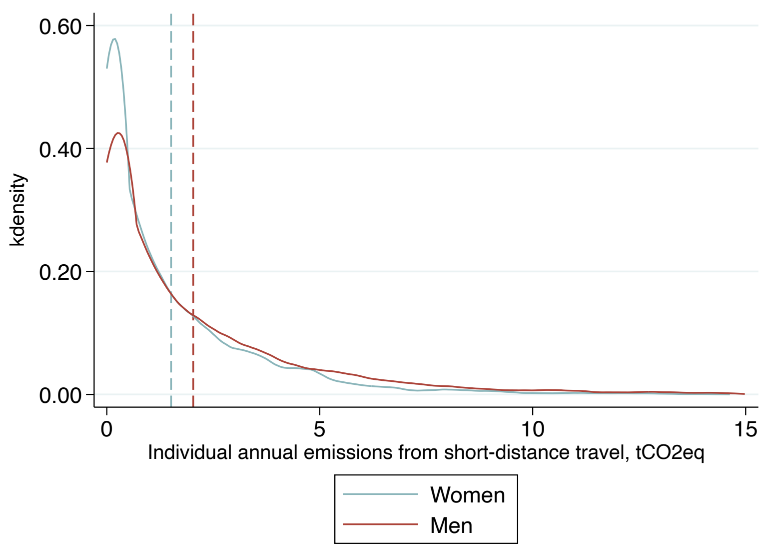
Notes: CO_2e emission intensities (in g CO_2e/km) for different transportation modes. Source: EMP trip-level data (N=44,759 for short-distance trips and N=30,938 for long-distance trips). Weighted averages across trips using the same transport mode category, using distances as weights.

Figure D.5: Distribution of Annual Carbon Footprints: Food.



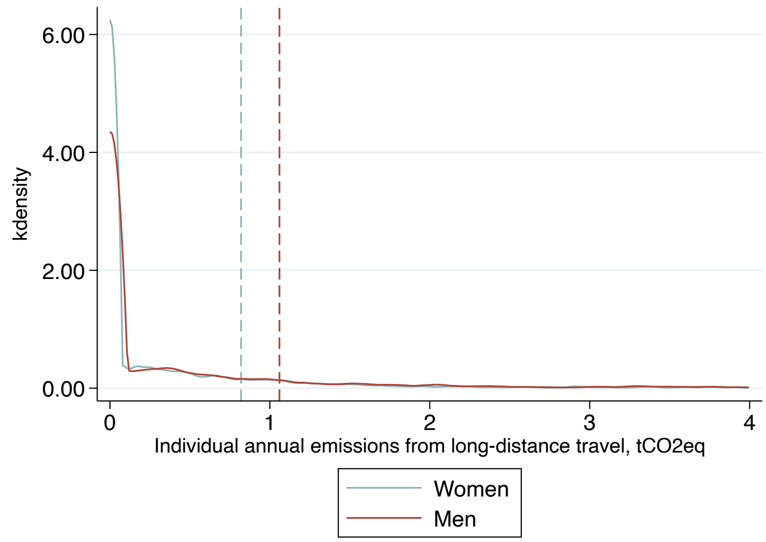
Notes: The blue dashed line indicates the average annual carbon footprints for women and the red dashed line the average annual carbon footprints for men, calculated with survey weights. Source: INCA3 (N=2,121).

Figure D.6: Distribution of Annual Carbon Footprints: Short-Distance Travel.



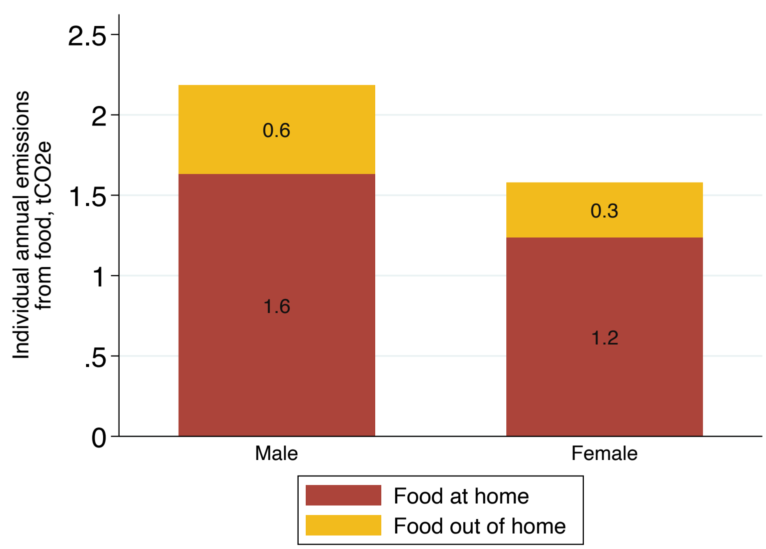
Notes: The blue dashed line indicates the average annual carbon footprints for women, and the red dashed line indicates the average annual carbon footprints for men, calculated with survey weights. Source: EMP (N=11,307).

Figure D.7: Distribution of Annual Carbon Footprints: Long-Distance Travel



Notes: The blue dashed line indicates the average annual carbon footprints for women and the red dashed line the average annual carbon footprints for men, calculated with survey weights. Source: EMP (N=11,307).

Figure D.8: Decomposition of Food Carbon Footprints Between Food at Home and Out of Home.

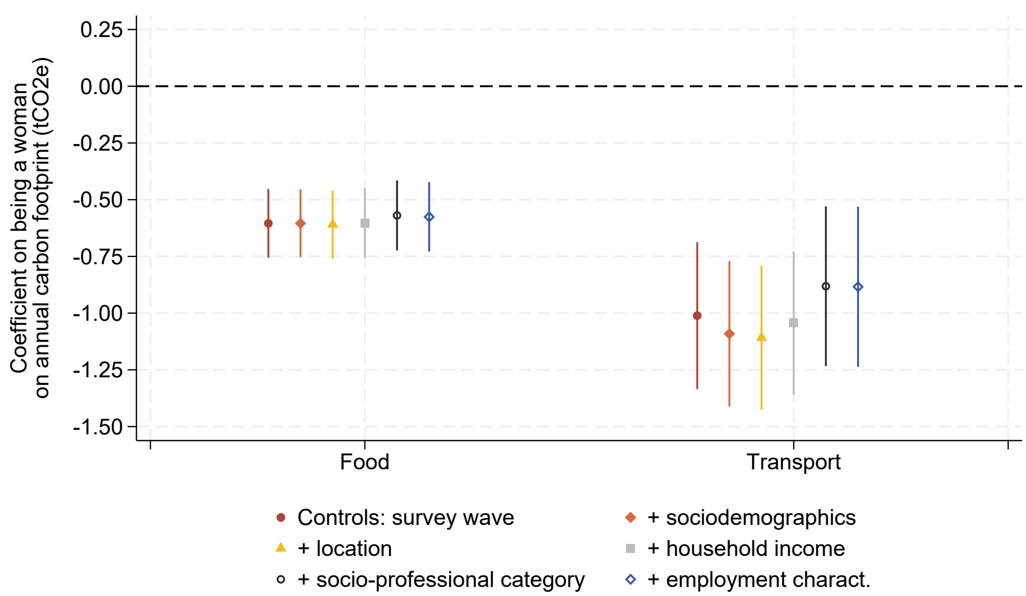


Notes: Food out-of-home includes food taken at friends' or relatives' home, on the workplace or at restaurants and take-aways. Source: INCA3 (N=2,121). Averages calculated with survey weights.

Figure D.9: Decomposition of Transport Carbon Footprints Between Work and Non-Work.

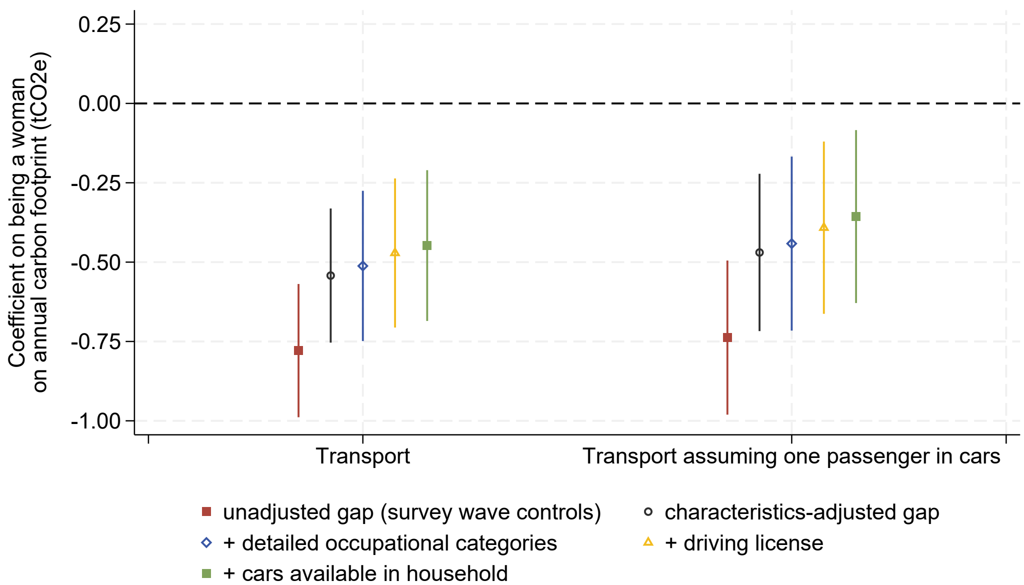
Notes: Work emissions are emissions from trips having a work-related purpose, either commuting or a business trip. Non-work emissions are emissions from trip having as purpose either leisure, shopping or escorting or another purpose. Source: transport: EMP (N=11,307). Averages calculated with survey weights.

Figure D.10: Conditional Gender Gap in Carbon Footprints, Sample of Individuals in Employment.



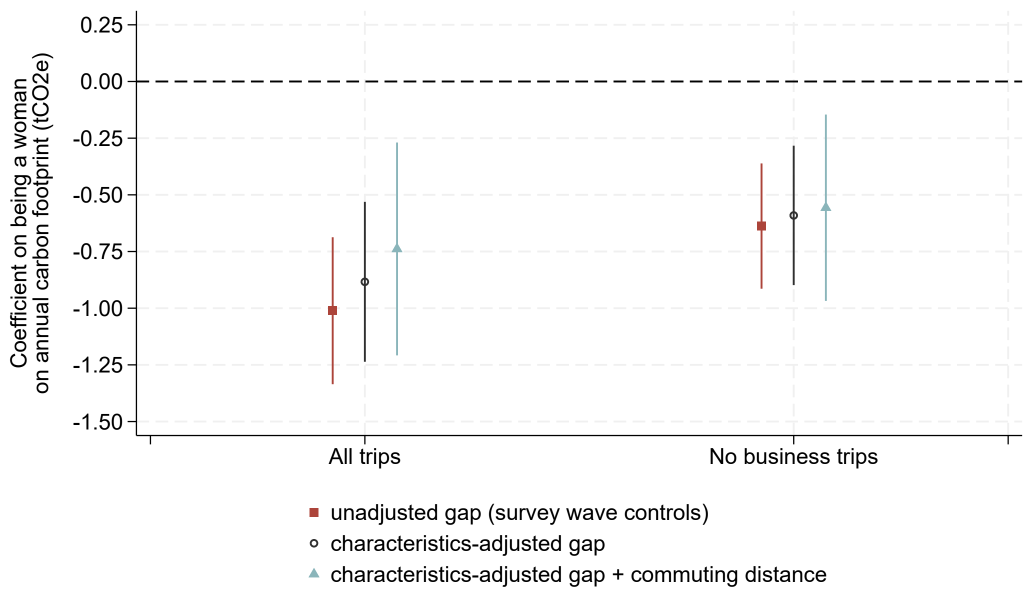
Notes: The point estimates and 95% confidence intervals show the estimated coefficient for the gender dummy "female" from separate OLS regressions, one for each consumption category, including an increasing number of control variables. "Controls: survey wave" only controls for the time of year when the survey is conducted. "+ sociodemographics" additionally controls for age, education and household size. "+location" additionally controls for rural vs. urban and size of the urban area of residence. "+household income" additionally controls for household income. "+socio-professional category" additionally controls for socio-professional category and employment status. "+employment charact." additionally includes an indicator variable for whether the individual has an atypical working time, defined as going to work or coming back from work between midnight and 5am, or going to work after 3:59pm. Source: Food consumption: INCA3 (N=1,142 for all employed); transport: EMP (N=5,631 for all employed).

Figure D.11: Conditional Gender Gap in Transport Carbon Footprint, additional controls and robustness checks.



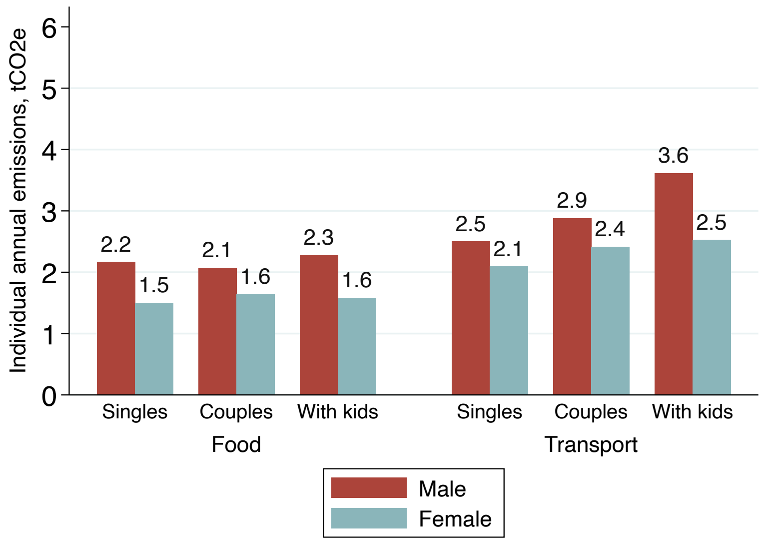
Notes: The point estimates and 95% confidence intervals show the estimated coefficient for the gender dummy "female" from separate OLS regressions, one for each measure of individual carbon footprint from transport, including an increasing number of control variables. "unadjusted gap (survey wave controls)" only controls for the time of year when the survey is conducted. "characteristics-adjusted gap" additionally controls for age, education and household size, rural vs. urban and size of the urban area of residence, household income, employment status and broad socio-professional category. "+ detailed occupational categories" replaces the five occupational categories with 42 categories mixing activity status (e.g., student or inactive outside the retired), detailed occupation category (30 categories) for the employed and unemployed, and broad occupation category for the retired. "+ driving license" additionally controls for whether the respondent has a driving license. "+ cars available in household" additionally controls for whether the household owns more than one car. The measure of carbon footprint in "Transport assuming one passenger in cars" does not divide car trips emissions by the number of passengers in the car for that trip. Source: transport: EMP (N=11,016).

Figure D.12: Gender Gap in Transport Carbon Footprints among the Employed, the role of Commuting and Business Trips.



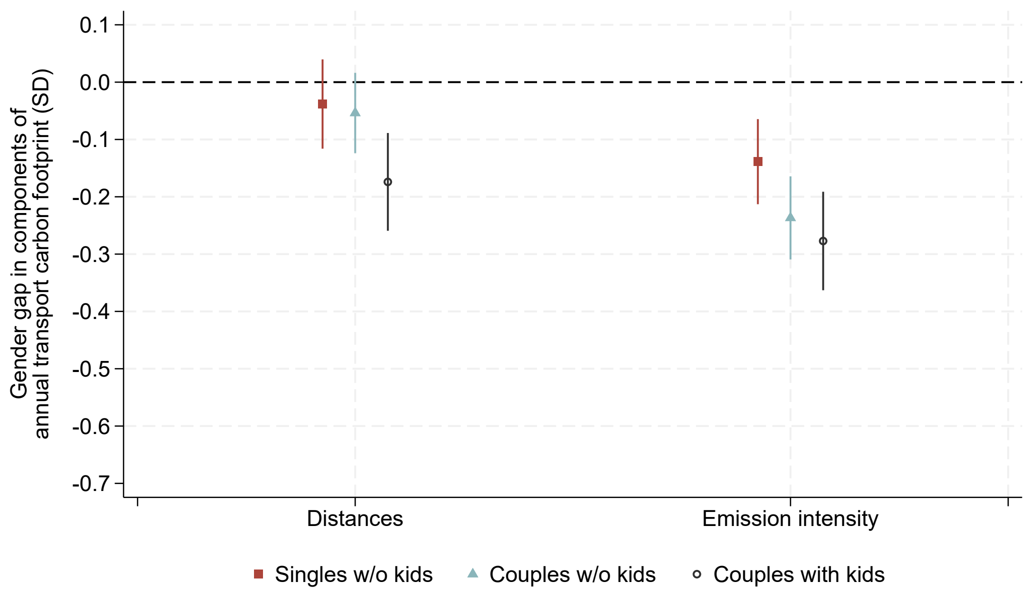
Notes: The point estimates and 95% confidence intervals show the estimated coefficient for the gender dummy "female" from separate OLS regressions, one for each scope of emissions "All emissions" measures individual carbon footprint from all short-distance and long-distance trips, while "No emissions from business trips" exclude work-related trip other than commuting from the calculation of individual footprint. The regressions include an increasing number of control variables. "unadjusted gap (survey wave controls)" only controls for the time of year when the survey is conducted. "characteristics-adjusted gap" additionally controls for age, education and household size, rural vs. urban and size of the urban area of residence, household income, employment status, broad socio-professional category, and night shift work. "characteristics-adjusted gap + commuting distance" also controls for commuting distance when available. Source: EMP (N=5,631 and N=2,610 for the regression with commuting distance).

Figure D.13: Individual Carbon Footprints from Annual Food Consumption and Transport Use by Gender and Household Composition.



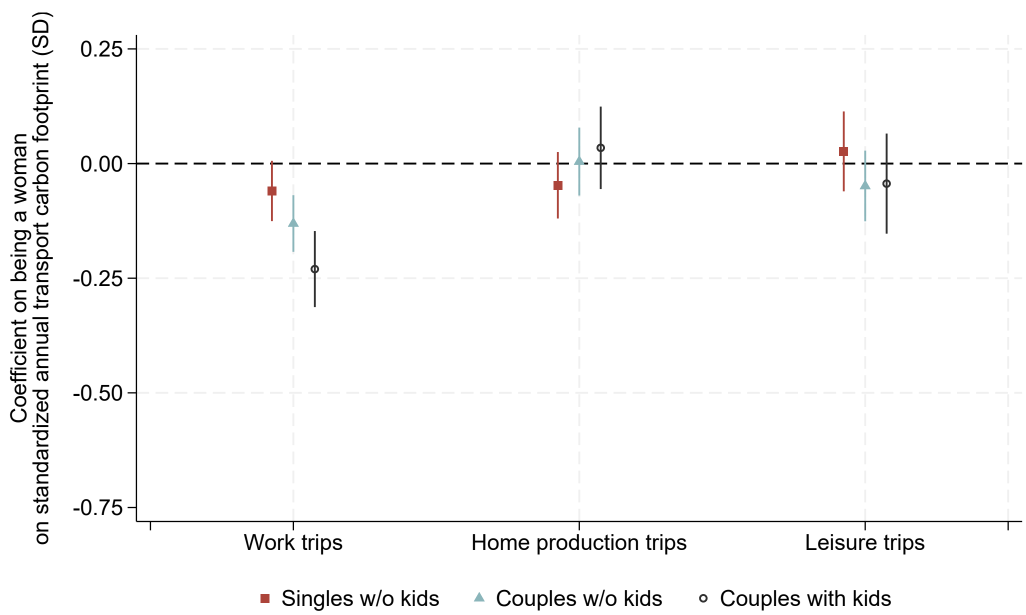
Notes: Source: Food consumption: INCA3 (singles without children: N=545; couples without children: N=721; couples with children: N=610); transport: EMP (singles without children: N=3,490; couples without children: N=3,723; couples with children: N=2,943).

Figure D.14: Characteristics-adjusted Components of the Gender Gap in Transport Carbon Footprints by Household Type.



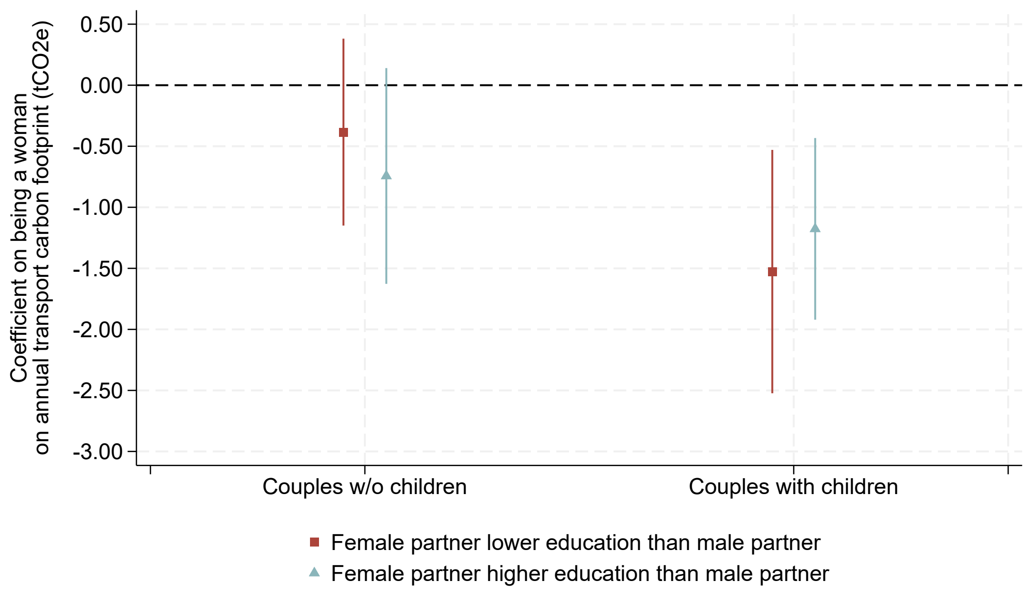
Notes: The point estimates and 95% confidence intervals in red show the estimated coefficient for the gender dummy "female" from separate OLS regressions using standardized distance traveled and standardized emission intensity (carbon footprint per kilometer traveled) as outcomes, controlling for the variables listed thereafter, for the subsample of single adults without children. The point estimates, and confidence intervals in blue and black show the corresponding coefficients for the subsample of individuals living in a couple without children and individuals living in a couple with children, respectively. Control variables include the time of year when the survey is conducted, age, education, household size (not for singles), rural vs. urban and size of the urban area of residence, household income, individual employment status and socio-professional category. Source: transport: EMP (singles without children: N=3,490; couples without children: N=3,723; couples with children: N=2,943).

Figure D.15: Characteristics-adjusted Gender Gap in Transport Carbon Footprints by Household Type, separating Emissions by Trip Purpose.



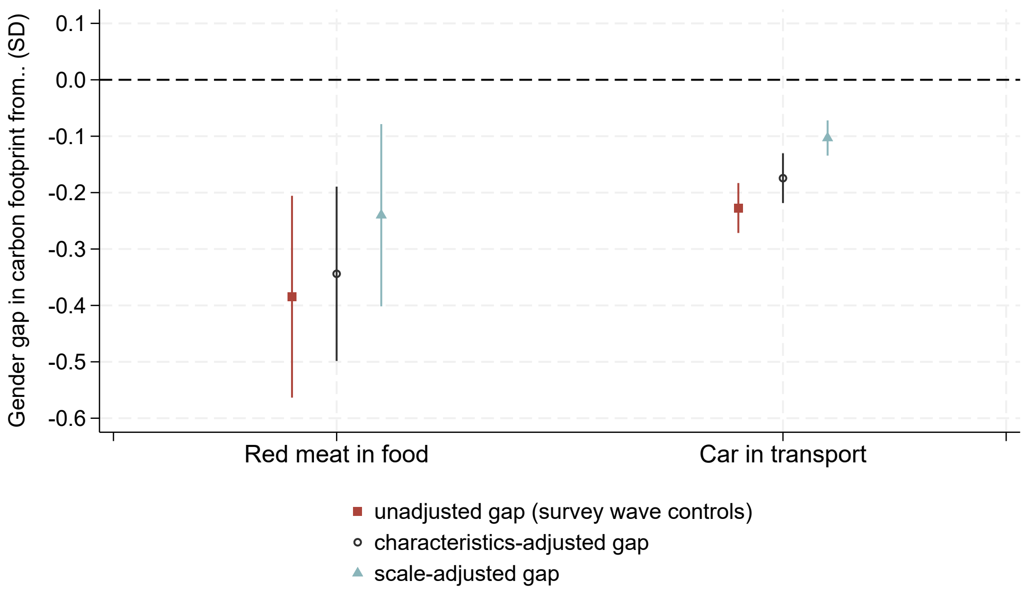
Notes: The point estimates and 95% confidence intervals in red show the estimated coefficient for the gender dummy "female" from separate OLS regressions, by trip purpose, controlling for the variables listed thereafter, for the subsample of single adults without children. The point estimates, and confidence intervals in blue and black show the corresponding coefficients for the subsample of individuals living in a couple without children and individuals living in a couple with children, respectively. Control variables include the time of year when the survey is conducted, age, education, household size (not for singles), rural vs. urban and size of the urban area of residence, household income, individual employment status and socio-professional category. Source: transport: EMP (singles without children: N=3,490; couples without children: N=3,723; couples with children: N=2,943).

Figure D.16: Characteristics-adjusted Gender Gap in Transport Carbon Footprints by Relative Educational Attainment in Household.



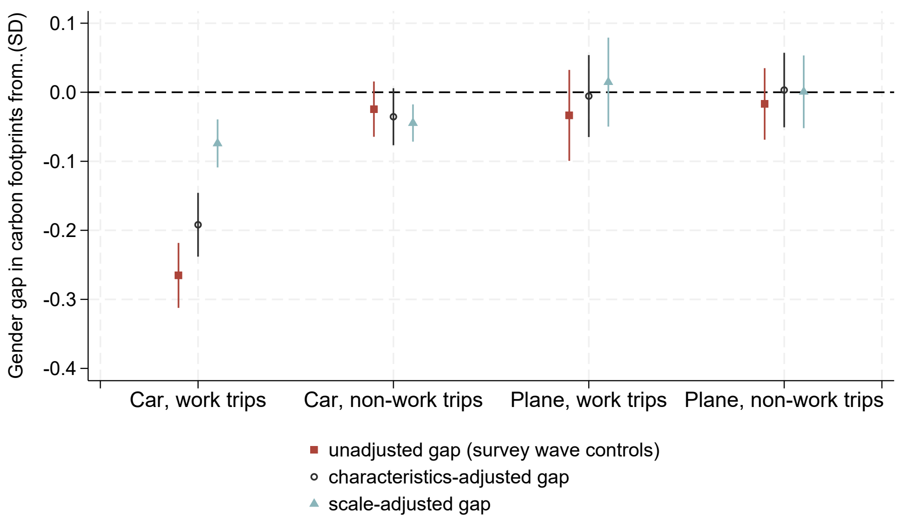
Notes: The point estimates and 95% confidence intervals in red show the estimated coefficient for the gender dummy "female" from separate OLS regressions, one for each household type subsample (couple without children and couple with children), controlling for the variables listed thereafter, for the subsample of individuals part of a couple where the female partner holds a strictly lower educational attainment than the male partner (with a hierarchical classification of highest diploma in eight categories). The point estimates, and confidence intervals in blue show the corresponding coefficients for the subsample of individuals part of a couple where the female partner holds a strictly higher educational attainment than the male partner. Control variables include the time of year when the survey is conducted, age, education, household size, rural vs. urban and size of the urban area of residence, household income, individual employment status and socio-professional category. Source: Transport: EMP (Couples without children: female partner lower education: N=1,156; female partner higher education: N=1,046. Couples with children: female partner lower education: N=645; female partner higher education: N=958).

Figure D.17: Gender Gap in Carbon Footprints from Red Meat and Car



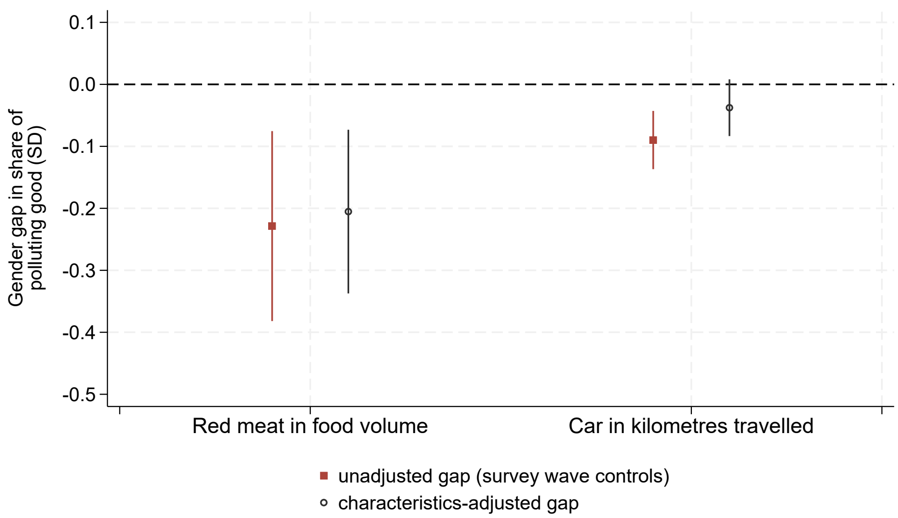
Notes: The point estimates and 95% confidence intervals in red show the estimated coefficient for the gender dummy "female" from separate OLS regressions of standardized emissions from red meat consumption and standardized emissions from car trips. The point estimates and 95% confidence intervals in blue show the estimated coefficient for the gender dummy "female". "unadjusted gap (survey wave controls)" only controls for the time of year when the survey is conducted. "characteristics-adjusted gap" additionally controls for age, education, household size, rural vs. urban and size of the urban area of residence, household income and employment status and socio-professional category. "scale-adjusted gap" additionally controls for reported daily caloric intake for the regression on food and for reported total distances traveled for the regression on transport. Sources: INCA3 (N=1,957); EMP (N=11,016).

Figure D.18: Gender Gap in Car versus Plane Emissions.



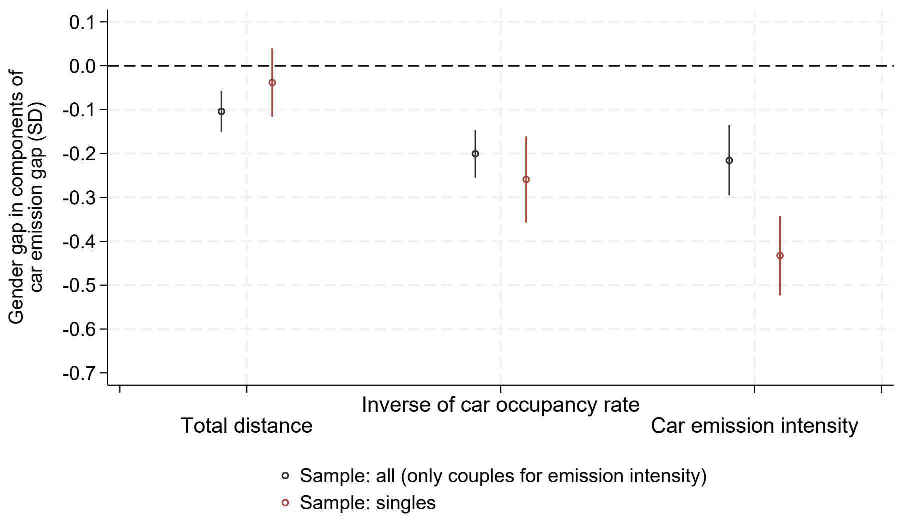
Notes: The point estimates and 95% confidence intervals in blue show the estimated coefficient for the gender dummy "female" from separate OLS regressions of standardized emissions from car work trips, car non-work trips, plane work trips and plane non-work trips. "unadjusted gap (survey wave controls)" only controls for the time of year when the survey is conducted. "characteristics-adjusted gap" additionally controls for age, education and household size, rural vs. urban and size of the urban area of residence, household income, employment status, broad socio-professional category. "scale-adjusted gap" additionally controls for total distances traveled for the trip purpose considered, in kilometers. Work trips include commuting trips and other business-related trips. Non-work trips include all other trips. Source: EMP (N=11,016).

Figure D.19: Gender Gap in the Share of Red Meat in Food Volumes and the Share of Car in Total Distances



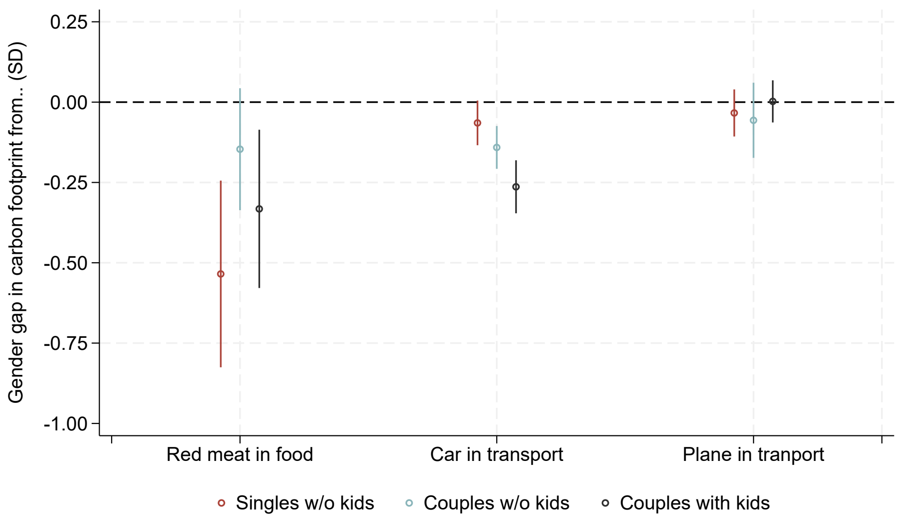
Notes: The point estimates and 95% confidence intervals show the estimated coefficient for the gender dummy "female" from separate OLS regressions of standardized share of red meat volume in total food volume, and share of kilometers by car in total kilometers traveled."unadjusted gap (survey wave controls)" only controls for the time of year when the survey is conducted. "characteristics-adjusted gap" additionally controls for age, education and household size, rural vs. urban and size of the urban area of residence, household income, employment status, broad socio-professional category. Source: INCA3 (N=1,957); EMP (N=10,967).

Figure D.20: Characteristics-Adjusted Gender Gap in Components of the Car Emission Gap.



Notes: The point estimates and 95% confidence intervals show the estimated coefficient for the gender dummy "female" from OLS regressions including the full set of control variables, one for each determinant of the gender car emissions gap, for the sample of all individuals (black) and for the subsample of singles (red). The first measure is the standardized total distance traveled with any transport modes, in kilometers; the second is the standardized inverse of the occupancy rate for car trips, for the subsample of individuals with at least one car trip; the third one is a standardized measure of the emission intensity of the car used for car trips, measured differently for all versus singles. The inverse occupancy rate is defined as one divided by the number of occupants per car. For the sample of all individuals, the emission intensity designates the share of car kilometers driven with the most carbon-intensive car of the household, restricting the analysis to couples owning at least two cars. For singles, it designates the emission intensity of the main car owned by the individual, measured in gCO_2e per kilometer. Source: EMP (all: N=11,016 for distance, N=7,998 for inverse occupancy rate, N=3,751 for emission intensity among couples with two cars; singles: N=3,490 for distance, N=2,230 for inverse occupancy rate, N=2,418 for own car emission intensity.)

Figure D.21: Characteristics-adjusted Gender Gap in Carbon Footprints from Meat, Car and Plane by Household Type



Notes: The point estimates and 95% confidence intervals show the estimated coefficient for the gender dummy "female" from separate OLS regressions, one for each consumption category, by household arrangement : single individuals without children, couples without children, and couples with children. Control variables include the time of year when the survey is conducted, age, education, household size, rural vs. urban and size of the urban area of residence, household income, individual employment status and socio-professional category. Source: Food: INCA3 (singles without children: N=545; couples without children: N=721; couples with children: N=610); transport: EMP (singles without children: N=3,490; couples without children: N=3,723; couples with children: N=2,943).

D.2 Tables

Table D.1: Summary Statistics: Sociodemographics by Sample

	Food		Transport	
	Mean	**SD**	**Mean**	**SD**
Household size	2.74	1.39	2.67	1.33
Gender = Female (share)	0.52	0.50	0.52	0.50
Age (shares)				
18-44	0.45	-	0.44	-
45-64	0.37	-	0.36	-
65-79	0.17	-	0.20	-
Work Status (shares)				
Employed	0.57	-	0.55	-
Unemployed	0.09	-	0.07	-
Pupil/Student	0.05	-	0.05	-
Pensioner or inactive	0.23	-	0.27	-
Other inactive	0.06	-	0.05	-
Education (shares)				
Less than secondary or vocational degree	0.51	-	0.46	-
End of high school diploma	0.15	-	0.21	-
Higher education degree 3 years† or below	0.17	-	0.13	-
Higher education degree above 3 years†	0.17	-	0.20	-
Observations	1,957		11,016	

Notes: Summary statistics by sample for the main comparable sociodemographic variables. †:2 years for Transport survey. Income is not included because the definition differs widely across samples. In the Transport survey, income is defined by decile, while it is interval-coded in the Food survey. Sample weights are applied.

Table D.2: Average Carbon Footprints for Food and Transport, full sample vs trimmed sample excluding top and bottom 5 percent.

	Gender	N	Mean	Std. Dev.	Min	Max	Absolute Gap	Relative Gap
Panel A: Food								
Full sample	Men	887	2.19	1.00	0.36	9.60		
	Women	1234	1.58	0.63	0.22	4.65	-0.61	-27.9%
Trimmed	Men	799	2.04	0.64	1.03	3.71		
	Women	1112	1.53	0.45	0.79	2.75	-0.51	-25.0%
Panel B: Transport								
Full sample	Men	5202	3.10	4.72	0.00	55.72		
	Women	6067	2.33	4.00	0.00	75.86	-0.77	-24.8%
Trimmed	Men	4942	2.24	2.33	0.00	10.70		
	Women	5764	1.68	1.79	0.00	8.10	-0.56	-25.0%

Notes: Trimmed indicates that the bottom and top 5 per cent observations for each group separately, i.e. by gender, have been excluded. Units are in metric tons of CO_2 per individual per year. The absolute gap is the difference in mean between women's and men's footprint. The relative gap is the absolute gap divided by men's footprint.

Table D.3: Gelbach Decomposition for Food.

	Magnitude in tCO_2 (SE)	Gender gap in %
Unadjusted Gap	-0.615***	27.7
	(0.067)	
Adjusted with controls (Figure 2a, left panel)		
Characteristics-adjusted Gap	-0.534***	24.4
	(0.057)	
Total change (Unadjusted - Adjusted)	-0.081**	
	(0.039)	
Of which:		
Survey Wave	-0.013	
	(0.010)	
Sociodemographics	-0.002	
	(0.007)	
Location	-0.004	
	(0.007)	
Household Income	-0.004	
	(0.013)	
Employment Status & socio-professional category	-0.057**	
	(0.028)	
Same controls + calories (Figure 2b, left panel)		
Scale-adjusted Gap	-0.147***	6.7
	(0.051)	
Total change (Unadjusted - Adjusted)	-0.468***	
	(0.063)	
Of which:		
Survey Wave	-0.002	
	(0.005)	
Sociodemographics	-0.001	
	(0.008)	
Location	-0.001	
	(0.004)	
Household Income	-0.004	
	(0.012)	
Employment Status & socio-professional category	-0.032	
	(0.022)	
Calories	-0.428***	
	(0.050)	

Notes: Based on regressions with robust estimator. The "Unadjusted Gap" presents the result of the regression in which there are no controls (the only explanatory variable is the indicator for gender). The upper panel presents the decomposition of a model in which five groups of covariates are included. The lower panel presents the decomposition of a model in which six groups of covariates are included, the last control added represents the distance traveled by the individual. The gender gap in percent is calculated by dividing the absolute value of the difference in emissions between men and women by the average carbon footprint of men. The list of control variables is presented in Appendix B.

Table D.4: Gelbach Decomposition for Transport.

	Magnitude in tCO_2 (SE)	Gender gap in %
Unadjusted Gap	-0.769***	24.8
	(0.107)	
Adjusted with controls (Figure 2a, right panel)		
Characteristics-adjusted Gap	-0.542***	17.5
	(0.108)	
Total change (Unadjusted - Adjusted)	-0.233***	
	(0.050)	
Of which:		
Survey Wave	0.009	
	(0.010)	
Sociodemographics	-0.011	
	(0.013)	
Location	0.036***	
	(0.011)	
Household Income	-0.094***	
	(0.019)	
Employment Status & socio-professional category	-0.172***	
	(0.039)	
Same controls + distance (Figure 2b, right panel)		
Scale-adjusted Gap	-0.274***	8.8
	(0.088)	
Total change (Unadjusted - Adjusted)	-0.501***	
	(0.072)	
Of which:		
Survey Wave	0.006	
	(0.007)	
Socioddemographics	0.001	
	(0.005)	
Location	0.025***	
	(0.009)	
Household Income	-0.049***	
	(0.013)	
Employment Status & socio-professional category	-0.103***	
	(0.031)	
Distance	-0.382***	
	(0.060)	

Notes: Based on regressions with robust estimator. The "Unadjusted Gap" presents the result of the regression in which there are no controls (the only explanatory variable is the indicator for gender). The upper panel presents the decomposition of a model in which five groups of covariates are included. The lower panel presents the decomposition of a model in which six groups of covariates are included, the last control added represents the distance traveled by the individual. The gender gap in percent is calculated by dividing the absolute value of the difference in emissions between men and women by the average carbon footprint of men. The list of control variables is presented in Appendix B.

Table D.5: Predicted Food Carbon Footprints for Women in Different Household Arrangements.

	Predicted Mean Carbon Footprint	
	Unadjusted	Characteristics-adjusted
Single woman	1.492	1.523
(SE)	(0.003)	(0.009)
Woman in couple without children	1.632	1.612
(SE)	(0.002)	(0.012)
t-statistic	-43.313	-6.038
p-value	0.000	0.000

Notes: Comparison of predicted average annual food carbon footprints for single women vs women in a couple without children, based on a Welch's t-test. Unadjusted specifications include seasonal and weekday controls. Characteristics-adjusted specifications include seasonal and weekday, socio-demographic, location, income, and activity status controls.